



FACULTY OF COMPUTING

BCS2243

WEB ENGINEERING

SEMESTER II 2024/2025

TITLE : Fakulti Komputeran Petakom (MyPetakom)

SECTION : 01

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SITI NURIZZATI BINTI ZABIDI	CA23075	

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1.0 PRINCIPLE OF THE SOLUTION

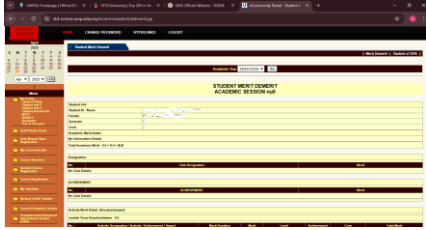
The principle of the solution is to create a web-based application, MyPetakom, for managing administrative tasks for the Persatuan Teknologi Komputer (Petakom), the student association at the Faculty of Computing, Universiti Malaysia Pahang Al Sultan Abdullah (UMPSA). Currently, many processes, such as membership registration, are performed manually. The proposed solution would automate these tasks and provide an integrated system for managing student membership, events, attendance, and merits.

MyPetakom will handle a few important tasks. First, it will allow users, such as students, staff and administrators to log in securely and access various functions based on their role. Students will be able to sign up for events, manage their profiles, and monitor their participation. Event advisers may organize and manage events, allocate areas to students, and monitor event operations. Administrators will be able to authorize student memberships, manage event data, and monitor the entire system.

The system will also make it easy to participate in the event. When students attend events, they will scan QR codes to check in, making it quick and easy to track who's present. The system will automatically assign merit points to each student based on their participation and attendance in events. Administrators will be able to view all the data, making it easy to track how much each student participates in the activities.

Overall, MyPetakom is designed to make managing Petakom activities easier for all participants. Students will easily be able to register for events, track their merit points, and manage their membership. Administrators will save time by using automated features to complete tasks like member approval and event management. The system will assist everyone in maintaining organization and guarantee the efficient operation of Petakom activities.

2.0 REVIEW OF EXISTING SYSTEMS

Existing System	 UMPSA Official Portal	 UMPSA E-community Portal
Description	This is the official portal for students at UMPSA, providing access to academic records, course registration, event participation, and some cocurricular activities. While it tracks academic progress and gives students access to university resources, its primary focus is academic related features.	This system is specifically designed to track merit points for students based on their participation in university events and activities. Students can check their accumulated merit points online, which are awarded for involvement in various activities like workshops, sports, or university functions.
Features	• Academic Record Management: Allow students to access their grades, course registrations and other academic data. • Event Registration: Students can register for events related to the university. • Co-Curricular Activity Tracking: Some tracking of student involvement in extracurricular	• Merit Point Tracking: Tracks students' involvement in events and rewards them with merit points. • Online Portal: Students can log in to view their merit points and participation history. • Focus on Co-curricular Achievements: The system emphasizes rewarding students for their

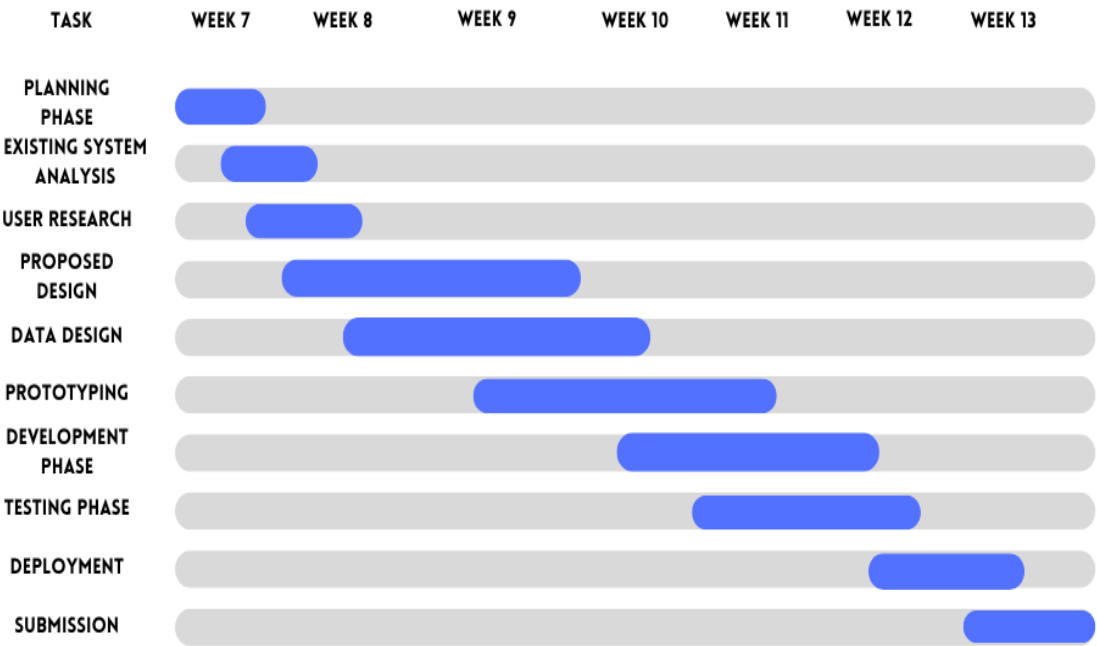
	activities, though is limited compared to other systems.	extracurricular involvement, encouraging further participation.
Advantages	• Centralized Access: Students have easy access to their academic records, courses, and event registration. • Easy Registration: Simple for students to register for university events. • Academic Integration: Strong focus on academic data and services.	• Simple Interface: It is easy for students to track their merits and see the activities they've participated in. • Transparent Merit Allocation: Students can clearly see how they earn merit points, making the process fair and accountable. • Focus on Co-Curricular Achievements: Rewards students for their participation in extracurricular activities.
Disadvantages	• Limited Co-curricular Management: Focuses mainly on academic records and lacks in-depth tracking of student participation in events and clubs. • No Merit or Rewards System: Does not have a merit-based system for student participation in non-academic activities. • Limited Event Management: Does not provide full event management or committee tracking capabilities.	• Limited Event Management: Does not allow students or administrators to create or manage events. • Lacks Committee Role Tracking: Does not track which students are involved in committees or leadership roles within events. • No Integration with Academic Records: Does not link co-curricular achievements with students' academic progress.

3.0 PROJECT MANAGEMENT FRAMEWORK

Project Initiation and Conception	The MyPetakom Web Application aims to digitalize administrative processes for Petakom, including membership registration, event management, attendance tracking, and merit calculation. The system will replace manual tasks, improving operational efficiency for students and staff at the Faculty of Computing, UMPSA.
Project Definition and Planning	The project will adopt the Agile Methodology. Agile is chosen because it supports fast-paced development, continuous feedback, and flexibility to adapt to stakeholder requirements throughout the project cycle. The project is divided into several modules, including Login and Membership Management, Event Management, Attendance Management, and Merit Management. The team will work concurrently on these modules through short iterative sprints to ensure faster delivery of functional components.
Project Launch and Execution	During the execution stage, the project team will use tools like Excel spreadsheets to track development progress and manage task assignments. Regular meetings will be held to synchronize team efforts, review sprint outcomes, and ensure that development aligns with project objectives.
Project Monitoring and Control	The project leader will actively monitor each member's performance and overall project progress. Sprint reviews will be conducted at the end of each iteration to identify what went well, what needs improvement, and whether any adjustments are required.

Project Closure	Upon completion of the MyPetakom Web Application, a final evaluation will be conducted to assess whether all project objectives have been successfully achieved. Comprehensive project documentation, including technical reports, user manuals, and deployment guides, will be prepared and submitted to stakeholders. Final deliverables will be handed over, and a post-project review will be organized to gather lessons learned for future improvements.
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4.0 PROJECT REQUIREMENTS

Non-functional requirements that can be considered for our project:

1. **Security:** The system should manage user sessions securely to prevent unauthorized access.
2. **Performance:** The system should load all web pages within three seconds under standard operating conditions.
3. **Usability:** The system should provide an intuitive and user-friendly interface that enables users to navigate easily without extensive technical knowledge.
4. **Scalability:** The system should be designed to scale efficiently, allowing the addition of new users, events, and attendance records without significant restructuring.
5. **Accessibility:** The system should provide alternative text for all critical images, such as event posters and QR codes.
6. **Data Integrity:** The system should ensure the accuracy, consistency, and integrity of user data, preventing data corruption or loss.
7. **Reliability:** The system should perform weekly automated backups to ensure data recovery in case of failure.

5.0 PROPOSED DESIGN

5.1 USE CASE DIAGRAM

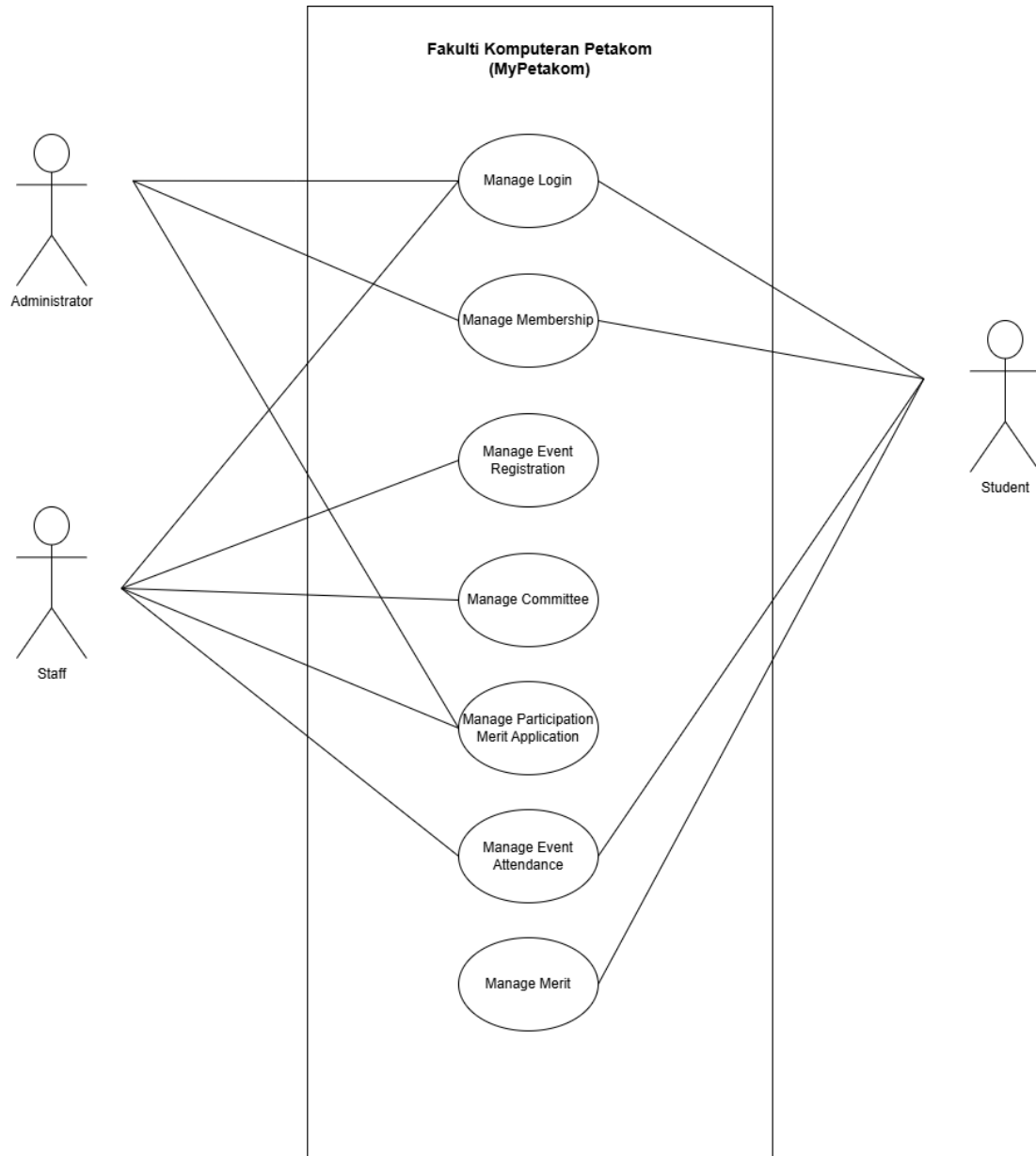


Figure 5.1 Use Case Diagram

5.2 USE CASE SCENARIOS

Use Case 1	Manage Login
Brief Description	This use case allows all registered users to login to the system.
Actors	• Staff (Petakom Advisor, Administrator) • Student
Preconditions	User has already registered to the system.
Basic Flow	1. The use case starts when the user clicks on the “Login” page. 2. The system displays the Login Page to the user. 3. The User will enter their username, password and their user type, which is either Administrator, Staff or Student. 4. The User will click on <<Login>> button. [A1 – Forgot Password] 5. The system displays the user’s dashboard. 6. The use case ends.
Alternative Flow	A1 – Forgot Password 1. The system displays an error message. 2. The User clicks on <<Forgot Password>> button. 3. The system will redirect User to Redeem Password page. 4. The User enters their email to redeem their password. 5. The system sends the User a temporary password through email. 6. The User goes through their email and redeem password. 7. Continue with step 5 from the basic flow.
Exception Flow	Not applicable.
Post Conditions	Users can access MyPetakom system.
Constraints	• The user must have a valid username, password, and email to log in.

Use Case 2	Manage Membership
Brief Description	This use case is to allow the Users to apply for membership and manage members' profiles.
Actors	• Student • Staff (Administrator)
Preconditions	Users have registered into the system.
Basic Flow	Student: 1. The use case starts when the student clicks on the “Membership Registration” page. 2. The system displays the Membership Registration page to the student. 3. The student will enter their username, email, password, select their role from the dropdown. 4. The student will log in and go to their profile. 5. The student uploads a valid student card in images. 6. The student will click on the <<Save>> button. [A1-Incomplete or Incorrect Information Submitted] 7. The system stores the registration request and marks the status as “Pending”. 8. The student can see the updated approval status on their dashboard after administrator verified the status. 9. The use case ends. Administrator: 1. The use case starts when the Administrator clicks on the “Manage Membership” section from their dashboard or admin panel. 2. The system displays the membership management interface to the Administrator, listing all members and their profile details. 3. The Administrator will select a member and make the necessary changes to their profile.

	4. The Administrator will click on the <<Approved>> or <<Rejected>> button to confirm the changes. 5. The system updates the student's status accordingly. 6. The use case ends.
Alternative Flow	A1 – Incomplete or Incorrect Information Submitted by Student 1. The system displays an error message indicating that the information provided by the student is incomplete or incorrect. 2. The student will review and correct the form based on the error message. 3. The student will click on the <<Save>> button again to resubmit the corrected form. 4. The system processes the corrected registration details and creates the student's profile. 5. The system displays a confirmation message that the registration was successful. 6. The use case continues with step 7 from the Student Basic Flow.
Exception Flow	Not applicable
Post Conditions	1. Student has successfully registered their membership and profile in the system. 2. Administrator can manage and update member profiles in the system.
Constraints	• The user must have a valid email address and all required documents to register successfully. • The system must let administrators easily manage user profiles.

Use Case 3	Manage Event Registration
Brief Description	This use case is to allow the User to manage event details.
Actors	Staff
Preconditions	Event to be created has already been approved by the superiors.
Basic Flow	- This use case starts when the Staff accesses Event Advisor Dashboard. - The system displays the events handled by the Staff. - The Staff clicks on <<View Details>> button to View Details of the event chosen. - The system redirects the Staff to the View Details Page. - The system displays the details of the event including the title, event date, event status, event location, event description and event QR. [A1-Edit Event] - The Staff can click on <<Generate Report>> button if they want to document the event. - The Staff can return to the dashboard by clicking on <<Return to Dashboard>> button. - The system will redirect the Staff to the dashboard again. - The Staff can click on <<Create New Event>> button to create a new event. - The system redirects the Staff to the Create New Event Page. - The system displays a form, asking for information about the event. - The Staff fills up the form. [E1 – Invalid Date and Time] - The Staff clicks on <<Save Changes>> button once finished filling up the form. - The system saves and displays the event at the dashboard. - The use case ends.
Alternative Flow	A1 – Edit Event

	1. The Staff clicks on <<Edit>> button to edit the details of the event. 2. The Staff clicks on <<Save Changes>> button after done editing the event. 3. The system saves the changes made. 4. Continue with step 5 of the Basic Flow
Exception Flow	E1 – Invalid Date and Time 1. The Staff enters a date and time that overlaps with another event done by Petakom. 2. The system displays an error message at the form box for Event Date. 3. The Staff will have to enter a new date. 4. Continue with step 12 of the Basic Flow.
Post Conditions	The events created are listed in the system.
Constraints	• Staff can create new event only after they have gotten an approval letter.

Use Case 4	Manage Committee
Brief Description	This use case allows the User to assign members as committees for events.
Actors	Staff
Preconditions	• Event must already be created. • Students must be a member of Petakom.
Basic Flow	1. The use case starts when the Staff have accessed View Details Page. 2. The Staff clicks on <<View Committee>> button to view the students that have been assigned to the committee. 3. The system redirects the Staff to the View Committee Page.

	- The system displays information about the students that are a part of the committee, including their names, student ids and their committee roles. [A1-Remove Student] - The Staff can add new students to the committee by clicking on the <<Add>> button. - The system will redirect the Staff to the Assign Committee Page. - The system displays a form to add the new students. - The Staff fills up the form with the student's id, name and their committee role. - The Staff clicks on <<Save>> button once they are done with the form. - The system saves the student's data. - The use case ends.
Alternative Flow	A1 – Remove Student - The Staff clicks on <<Remove>> button to remove the student from the committee. - The system automatically saves the changes made. - Continue with step 4 from the Basic Flow.
Exception Flow	Not applicable.
Post Conditions	Committee members are recorded.
Constraints	Only members of Petakom can be added into a committee.

Use Case 5	Manage Participation Merit Application
Brief Description	This use case allows User to apply for student participation merit.
Actors	Staff
Preconditions	- The Event must be approved. - The Students in the committee are finalized.
Basic Flow	The use case starts when the Staff access to View Details Page 2.

	2. The Staff clicks on the <<Apply Participation Merit>> button to apply for the student participation merit. 3. The system redirects the Staff to the Apply Participation Merit Page. 4. The system displays a Participation Merit form. 5. The Staff fills up the form, including the event approval letter, number of students in committee, and the list of students in the committee. 6. The Staff clicks on <<Apply>> button once they are done filling up the form. 7. The system saves and automatically redirects the Staff to the View Details Page 1. 8. The system displays the event status as “Pending”. 9. The use case ends.
Alternative Flow	Not applicable.
Exception Flow	Not applicable.
Post Conditions	The system will display whether the application for Student Participation Merit is approved, pending or rejected.
Constraints	Only valid documents are accepted by the system.

Use Case 6	Manage Event Attendance
Brief Description	This use case allows Users to manage attendance slots and participants' check in.
Actors	• Staff • Student
Preconditions	• The Staff must be successfully logged into the system. • The Staff must have access rights to create attendance forms and view event forms.

	Students must have a device capable of scanning QR codes
Basic Flow	Staff - The use case starts when the staff clicks on the “View Attendance ” page. - The system displays the existing attendance slots to the staff. - The staff will click on the << + create>> button. - The staff will select the event, date, time and search the location for the event that will be held.[A1-Incomplete Information Submitted] - The staff click <<submit>> button when finish fill the form. - The staff can click << update >> button to update the current information about the event attendance slot. - The staff will click on <<View Details>> button. - The system displays the details about the event and the attendance slot QR. - The use case ends. Student - The student scan the QR code. - The system displays the Attendance Form Page to the student. - The student enter student ID and password. - The student clicks on <<verify>> button. - The use case end.
Alternative Flow	[A1-Incomplete Information Submitted] - The system displays an error message indicating that the information provided in the event form is incomplete or incorrect. - The user reviews and corrects the form based on the error message.

	3. The user clicks on the <<submit>> button again to resubmit the corrected event form. 4. The system processes the corrected event details and saves the event information. 5. The system displays a confirmation message that the event has been successfully created. 6. The use case continues with step 5 from the Basic Flow.
Exception Flow	No applicable
Post Conditions	• Staff can view the list of respondents (students who have checked in). • Student attendance is successfully recorded in the system under the correct event/form.
Constraints	• Staff must fill in all required fields (event, date, time and search the location) before creating the Attendance Form. • QR code must be generated immediately after Attendance Slot Form creation without delay. • Staff can only view event forms and respondents for events they created. • Students must scan a valid and active QR code to access the Attendance Form. • Students must correctly enter their student ID and password to successfully check in. • Each student is allowed to check-in only once per event.

Use Case 7	Manage Merit
Brief Description	This use case allow Users to claim missing merit and display the current cumulative merits.

Actors	Student
Preconditions	Staff have applied for student participation merit.
Basic Flow	- The use starts when the student have accessed the Merit Summary Page. - The system will display the Missing Merit Awarded page contains Student ID, Event ID, Event Title, Role, Merit Score. [A1: Missing Merit] - The student clicks the <<QR>> button. - The system displays the QR page that contains the QR to view your merit profile. - After the student scanned the QR code, the system will display the Student Merit Details page. - The student clicks the <<Print as PDF>> button to save or print the document.
Alternative Flow	[A1: Missing Merit] - The student clicks the <<Merit Application>> button at navigation bar. - The system will display the Missing Merit Application page. - The student choose the Event Title, Role and Upload the Supporting Letter. - The student clicks <<Submit>> button . - Then, the student clicks the <<View Application>> button to view the apply missing merit. - The system displays the applyMissing Merit Application page.
Exception Flow	Not applicable
Post Conditions	Student's merit records are updated.
Constraints	- Students can only update or delete a missing merit before it is submitted. - Each merit claim must include a valid official participation letter only.

5.3 HYPERTEXT MODEL

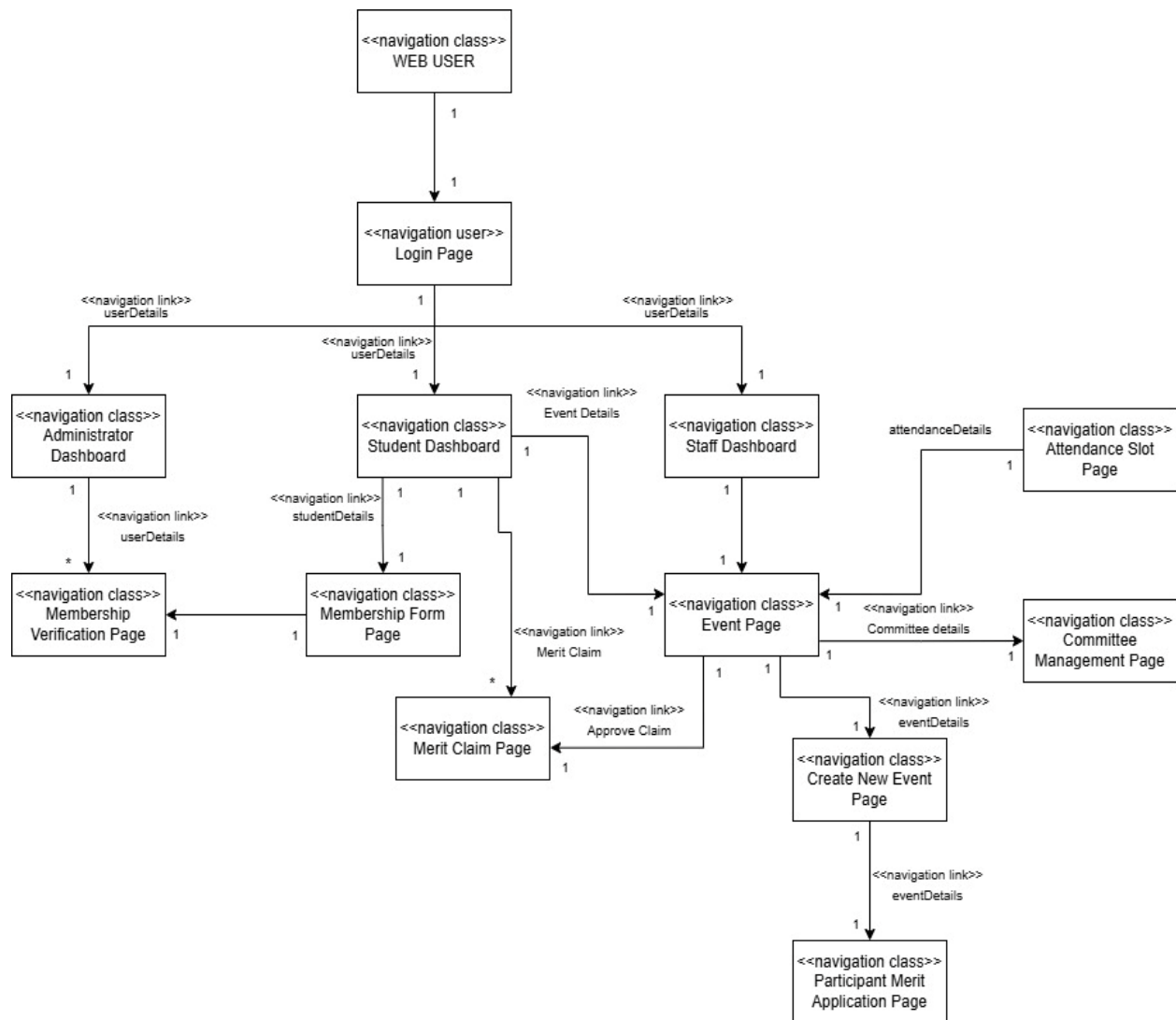


Figure 5.2 Hypertext Model

5.4 ACCESS MODEL

5.4.1 ADMINISTRATOR

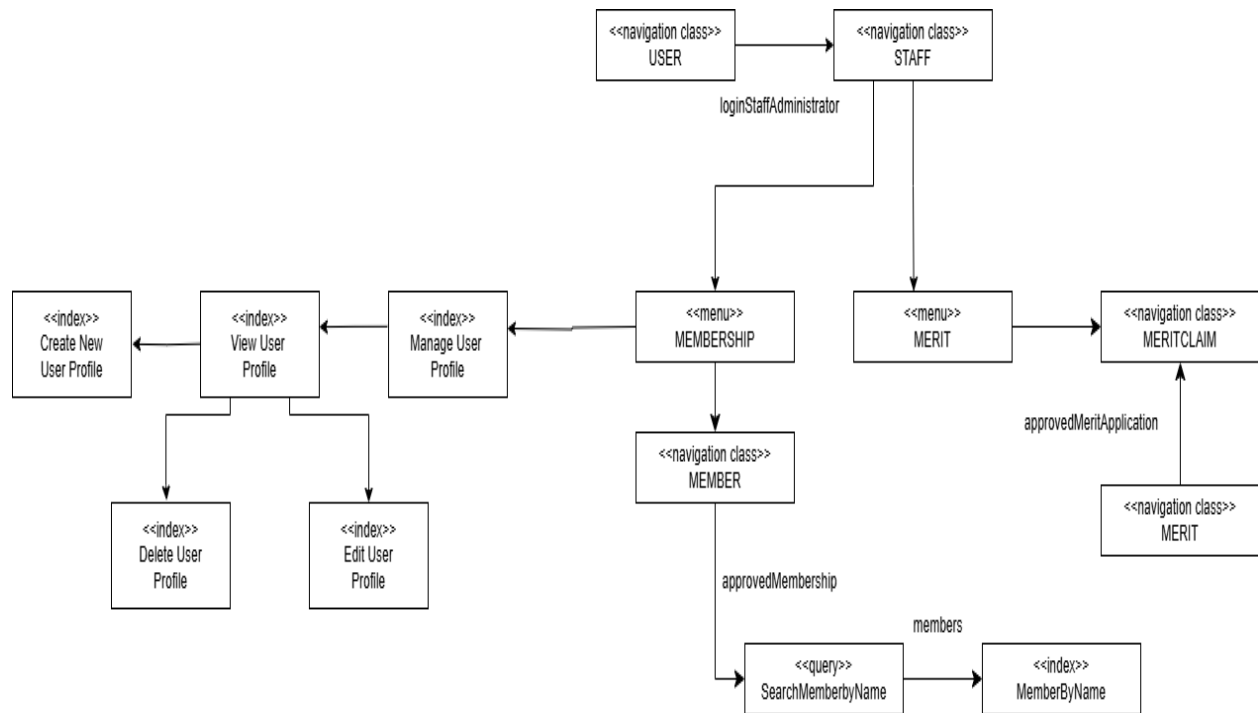


Figure 5.3 Access Model for Administrator

5.4.2 STUDENT

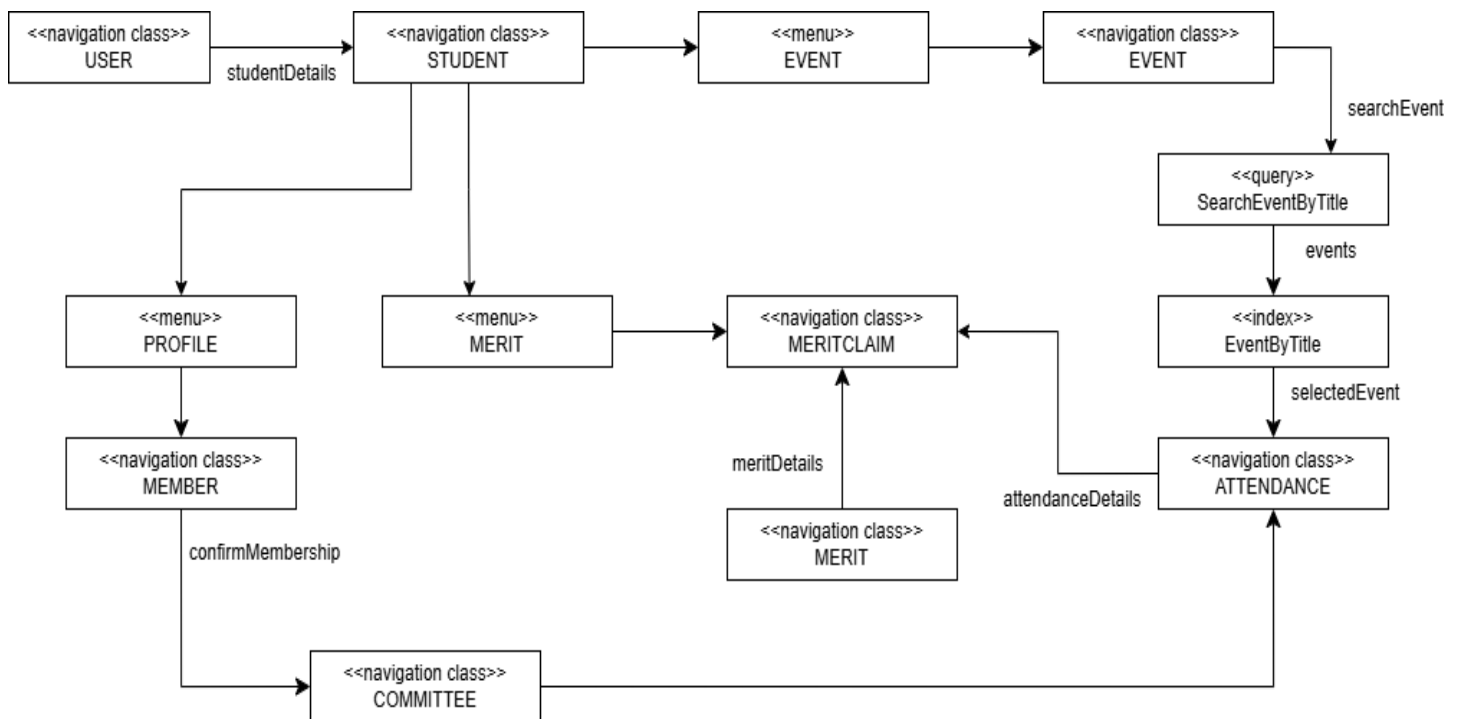


Figure 5.4 Access Model for Student

5.4.3 STAFF

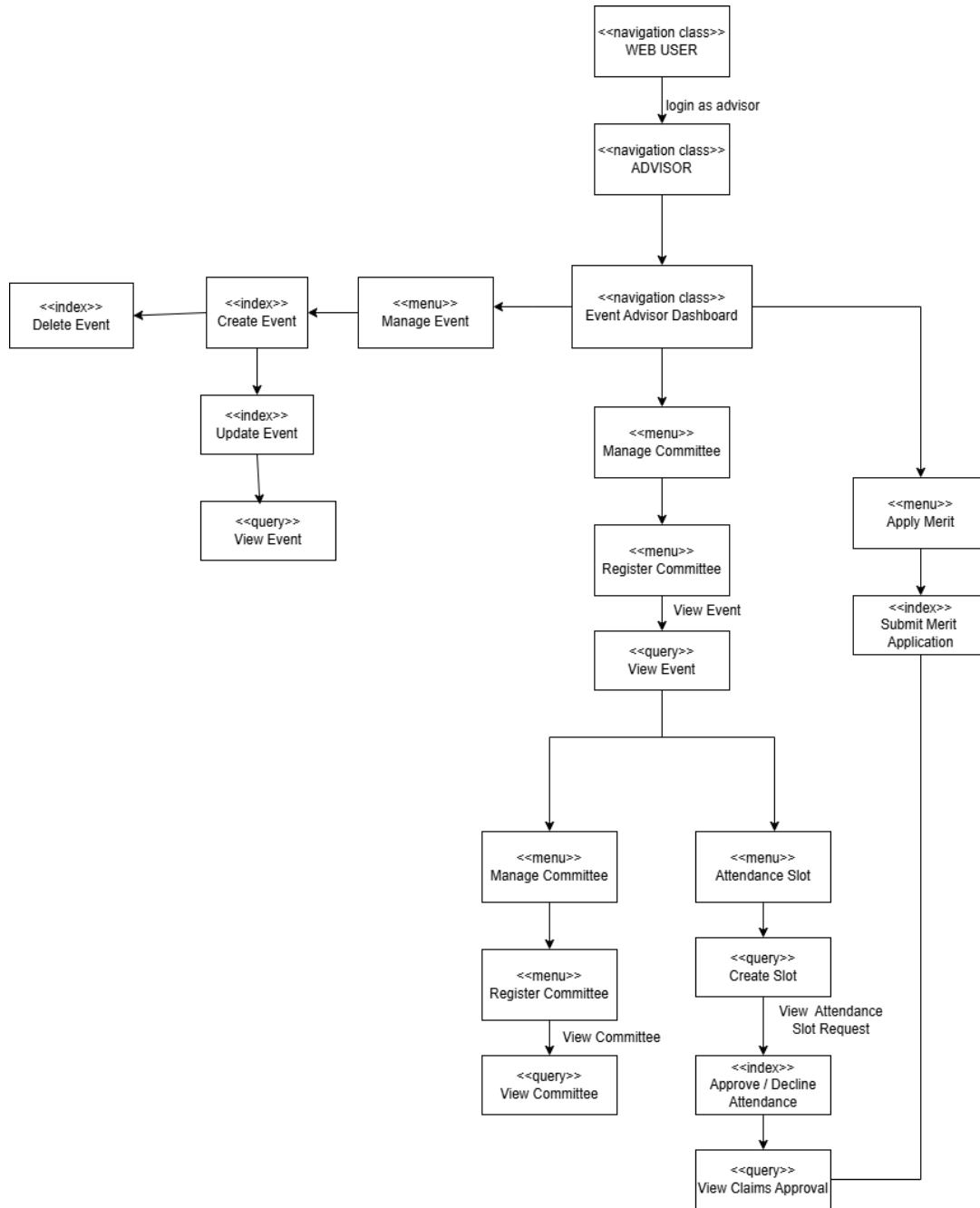
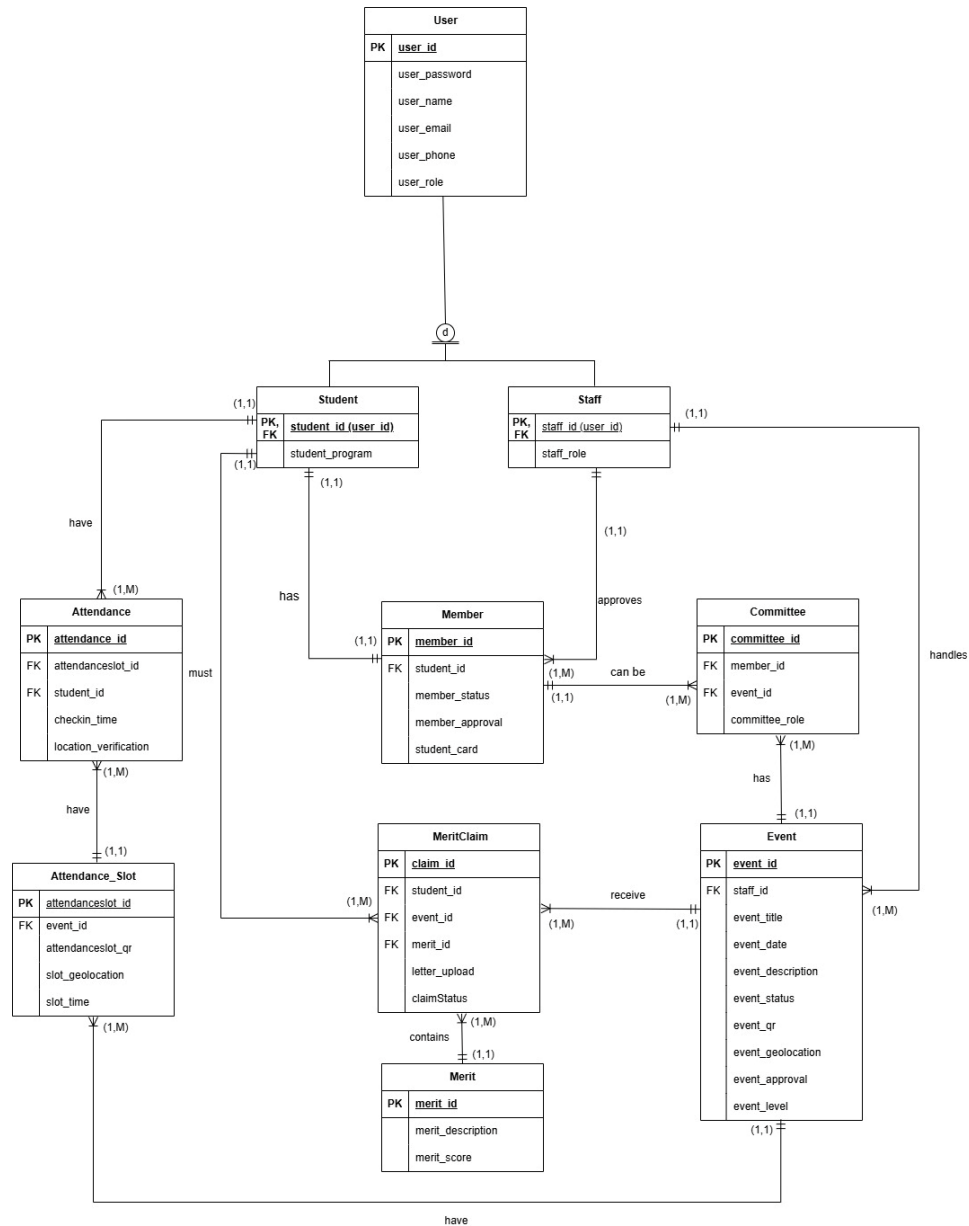


Figure 5.5 Access Model for Staff

6.0 DATA DESIGN

6.1 ENTITY RELATIONSHIP DIAGRAM (ERD)



6.2 DATA DICTIONARY

6.2.1 User

Field Name	Description	Data Type	Constraint
user_id	Unique ID for each user of the system	VARCHAR(10)	PK
user_password	Password set by the user	VARCHAR(50)	
user_name	Name of the user	VARCHAR(100)	
user_email	Email of the user	VARCHAR(100)	
user_phone	Phone number of the user	VARCHAR(12)	
user_role	Role of user	VARCHAR(30)	

6.2.2 Staff

Field Name	Description	Data Type	Constraint
staff_id	Unique ID for the staff	VARCHAR(10)	PK, FK
staff_role	Role representing the user	VARCHAR(50)	

6.2.3 Student

Field Name	Description	Data Type	Constraint
student_id	Unique ID for the student	VARCHAR(10)	PK, FK
student_program	Program of the student	VARCHAR(50)	

6.2.4 Member

Field Name	Description	Data Type	Constraint
member_id	Unique ID for the members of Petakom	INT(5)	PK
student_id	ID representing the student	VARCHAR(10)	FK
member_status	Status of the member	VARCHAR(50)	
member_approval	Member approval status by which administartor	VARCHAR(10)	
student_card	Student card uploaded by student	BLOB	

6.2.5 Committee

Field Name	Description	Data Type	Constraint
committee_id	Unique ID for the committee created	INT(10)	PK
member_id	ID representing the members of Petakom in the committee	INT(5)	FK
event_id	ID representing the event handled by the committee	VARCHAR(10)	FK
committee_role	Roles of committee	VARCHAR(50)	

6.2.6 Event

Field Name	Description	Data Type	Constraint
event_id	Unique ID for the event	VARCHAR(10)	PK
staff_id	ID representing the staff that handles the event	VARCHAR(10)	FK
event_title	Name of the event	VARCHAR(50)	
event_date	Date of the event occurring	DATE	
event_description	Description of the event	VARCHAR(200)	
event_status	Status of the merit application for the event	VARCHAR(50)	
event_qr	Event QR code	BLOB	
event_geolocation	Place where the event is being held	POINT	
event_approval	Files contain the official approval for the event	MEDIUMBLOB	
event_level	Type of the event	VARCHAR(20)	

6.2.7 Attendance_Slot

Field Name	Description	Data Type	Constraint
attendanceslot_id	Unique ID for the attendance slot	VARCHAR(10)	PK
event_id	ID representing the attending event	VARCHAR(10)	FK
attendanceslot_qr	Attendance QR code	MEDIUMBLOB	
slot_geolocation	Place where the attendance is being taken	POINT	
slot_time	The date and time of the attendance being taken	DATETIME	

6.2.8 Attendance

Field Name	Description	Data Type	Constraint
attendance_id	Unique ID for the attendance taken	VARCHAR(10)	PK
attendanceslot_id	ID representing the attendance slot	VARCHAR(10)	FK
student_id	ID representing the students attending the event	VARCHAR(7)	FK
checkin_time	Time the attendance taken	TIMESTAMP	
location_verification	Verifying whether location is correct or not	VARCHAR(5)	

6.2.9 MeritClaim

Field Name	Description	Data Type	Constraint
claim_id	Unique ID for the claim made by the students	VARCHAR(10)	PK
student_id	ID representing the student making the merit claim	VARCHAR(7)	FK
event_id	ID representing the event claimed by the student	VARCHAR(10)	FK
merit_id	ID represented by the merit being claimed by the student	VARCHAR(5)	FK
letter_upload	Status of the merit letter uploaded by the student	MEDIUMBLOB	
claimStatus	Status of the merit claim, such as approved or rejected	VARCHAR(20)	

6.2.10 Merit

Field Name	Description	Data Type	Constraint
merit_id	Unique ID of the merit being claimed	VARCHAR(5)	PK
merit_description	Brief description of the merit being claimed	VARCHAR(200)	
merit_score	Score earned by the merit	INT(3)	

6.3 SAMPLE DATA

6.3.1 User

user_id	user_password	user_name	user_email	user_phone	user_role
1001	siti123	SITI	siti@gmail.com	01121312440	student
1002	adlin321	ADLIN	adlin@gmail.com	0142727018	staff
1003	nick00	NICK	nick09@gmail.com	0172256484	administrator

6.3.2 Staff

staff_id	staff_role
S001	Staff (Petakom Advisor)
A001	Staff (Administrator)
A002	Staff (Administrator)

6.3.3 Student

student_id	student_program
CB23013	BCS - Bachelor of Computer Science (Software Engineering)
CA23075	BCN - Bachelor of Computer Science (Computer Systems & Networking)
CD21123	BCM - Bachelor of Computer Science (Multimedia Software)

6.3.4 Member

member_id	student_id	member_status	member_approval	student_card
14	CB23013	Approved	Approved	

15	CA23075	Pending	Pending	
16	CD21123	Rejected	Rejected	

6.3.5 Committee

committee_id	member_id	event_id	committee_role
C001	14	E001	Main Committee
C002	15	E002	Participant

6.3.6 Event

Event_id	staff_id	event_title	event_date	event_description	event_status	event_qr	event_location	event_approval	event_level
E001	S001	MBLL	10-05-2025	E-game challenges	Approved		3.7333, 103.1256	Mbll.pdf	UMPSA
E002	S002	Sukom	31-05-2025	Sukan Komputeran	Approved		3.7233, 104.1250	Sukom.pdf	Faculty

6.3.7 Attendance_Slot

attendance_slot_id	event_id	attendanceslot_qr	slot_geolocation	slot_time
AS001	E001		3.7333, 103.1256	10-05-2025 09:00:00
AS002	E002		3.7233, 104.1250	31-05-2025 11:00:00

6.3.8 Attendance

attendance_id	attendanceslot_id	student_id	checkin_time	location_verification
AT001	AS001	CB23013	10-05-2025 09:10:00	Yes
AT002	AS001	CA23075	10-05-2025 09:05:00	Yes
AT003	AS002	CD21123	31-05-2025 14:00:00	No

6.3.9 MeritClaim

claim_id	student_id	event_id	merit_id	letter_upload	claimStatus
MC001	CB23013	E001	M005	/uploads/letter_CB23013_E001.pdf	Approved
MC002	CA23075	E001	M006	/uploads/letter_CA23075_E001.pdf	Rejected
MC003	CD21123	E002	M015	/uploads/letter_CD21123_E002.pdf	Pending

6.3.10 Merit

merit_id	merit_description	merit_score
M001	Main Committee in International Level	100
M002	Committee in International Level	70
M003	Participant in International Level	50
M004	Main Committee in National Level	80
M005	Committee in National Level	50
M006	Participant in National Level	40
M007	Main Committee in State Level	60
M008	Committee in State Level	40
M009	Participant in State Level	30
M010	Main Committee in District Level	40
M011	Committee in District Level	30
M012	Participant in District Level	15
M013	Main Committee in UMPSA Level	30
M014	Committee in UMPSA Level	20
M015	Participant in UMPSA Level	5

7.0 PRESENTATION MODEL

7.1 Module 1 (Siti Nurizzati Binti Zabidi – CA23075)

<<background>>

<>
logo
Petakom

<<text>>

<<button>>
Log In

<<button>>
Sign Up

<<Page>>
Account Registration

<<text>> user ID

<<input text>> user ID

<<text>> Name

<<input text>> username

<<text>> Email Address

<<input text>> email address

<<text>> Phone number

<<input text>> phone number

<<text>> Password

<<text>> password

<<text>> User Role

<<dropdown>> User Role(Student)

<<text>> Student Program

<<dropdown>> Program

<<button>>
Register

Figure 7.1 Registration Page for Student

<<background>>

<>
logo
Petakom

<<text>>

<<button>>
Log In

<<button>>
Sign Up

<<Page>>
Account Registration

<<text>> user ID

<<input text>> user ID

<<text>> Name

<<input text>> username

<<text>> Email Address

<<input text>> email address

<<text>> Phone number

<<input text>> phone number

<<text>> Password

<<text>> password

<<text>> User Role

<<dropdown>> User Role

<<button>>
Register

Figure 7.2 Registration Page for Staff

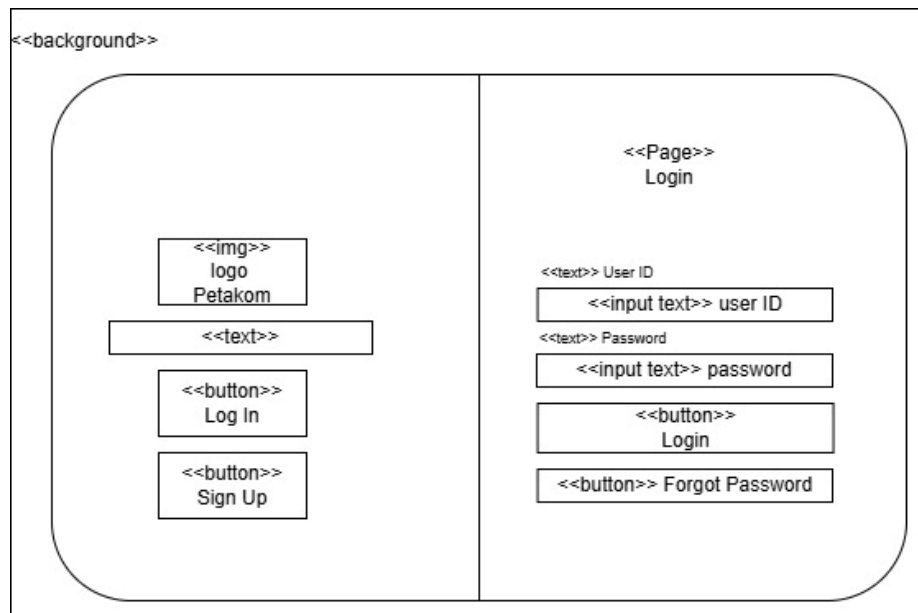


Figure 7.3 Login Page

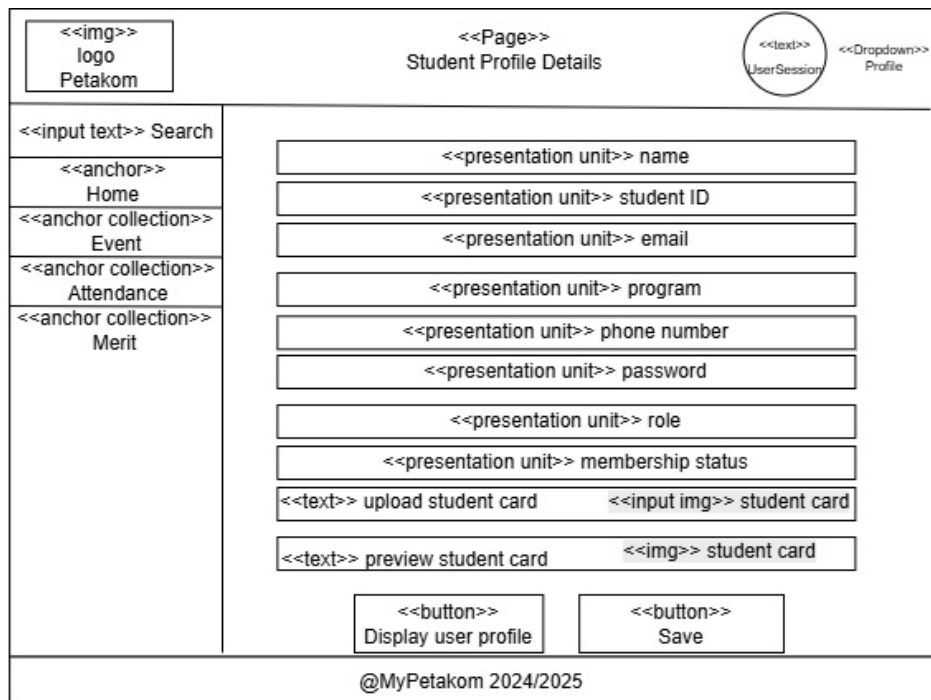


Figure 7.4 Student Profile Page

		Administrator Profile Details UserSession Profile	
Search Home Membership Event Merit		name staff ID email phone number password role	
@MyPetakom 2024/2025			

Figure 7.5 Administrator Profile Details Page

		Staff Profile Details UserSession Profile	
Search Home Event Attendance		name staff ID email phone number password role	
@MyPetakom 2024/2025			

Figure 7.6 Staff Profile Details Page

 UserSession Profile									
Search	Membership Management Search								
Home Membership Event Merit	<table border="1"> <tr> <td>Student ID</td> <td>Name</td> <td>Status</td> <td>Action</td> </tr> <tr> <td>Student ID</td> <td>Name</td> <td>Status</td> <td>Action</td> </tr> </table>	Student ID	Name	Status	Action	Student ID	Name	Status	Action
Student ID	Name	Status	Action						
Student ID	Name	Status	Action						
@MyPetakom 2024/2025									

Figure 7.7 Membership Management Page

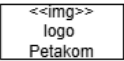
 UserSession Profile													
Search	User Profile Records Search Search by roles												
Home Membership Event Merit	<table border="1"> <tr> <td>User ID</td> <td>Name</td> <td>email</td> <td>phone number</td> <td>role</td> <td>Action</td> </tr> <tr> <td>Student ID</td> <td>Name</td> <td>Status</td> <td>phone number</td> <td>role</td> <td>Action</td> </tr> </table>	User ID	Name	email	phone number	role	Action	Student ID	Name	Status	phone number	role	Action
User ID	Name	email	phone number	role	Action								
Student ID	Name	Status	phone number	role	Action								
@MyPetakom 2024/2025													

Figure 7.8 List of User Page

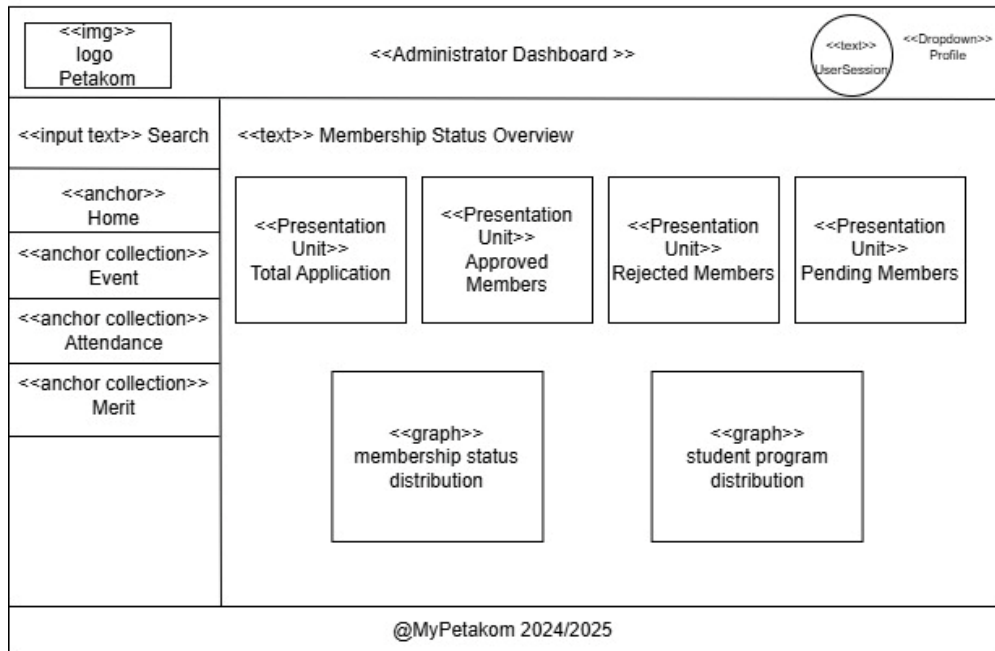


Figure 7.9 Administrator Dashboard

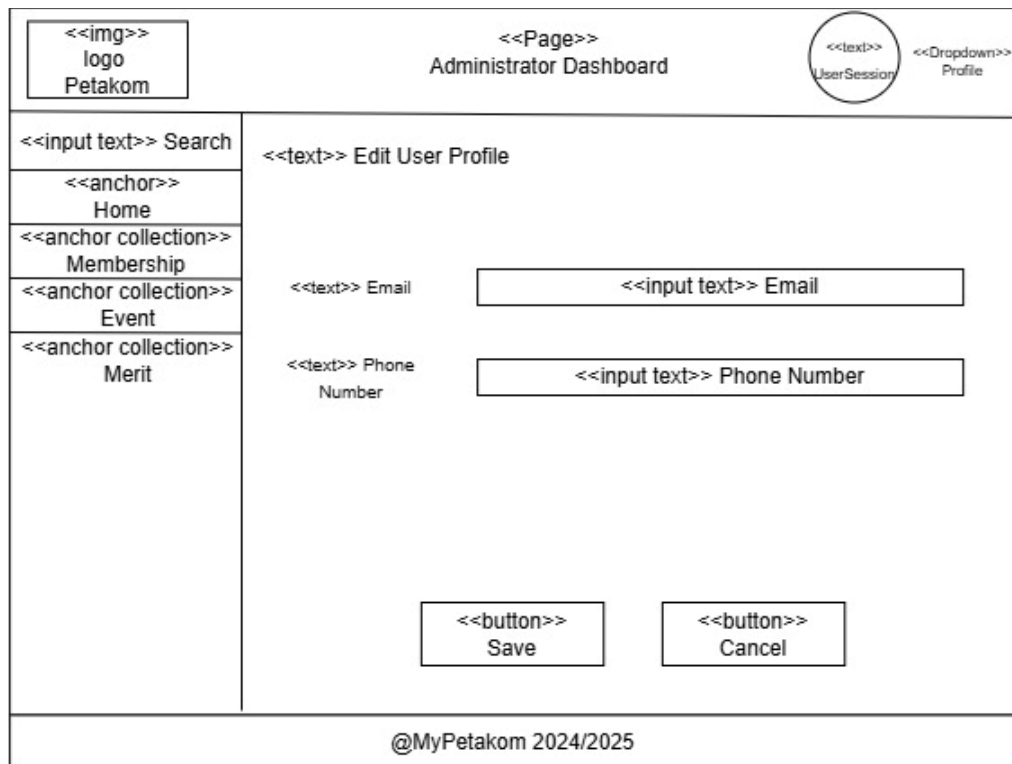


Figure 7.10 Edit User Profile Page

7.2 Module 2 (Siti Najwa Binti Bakri – CB23013)

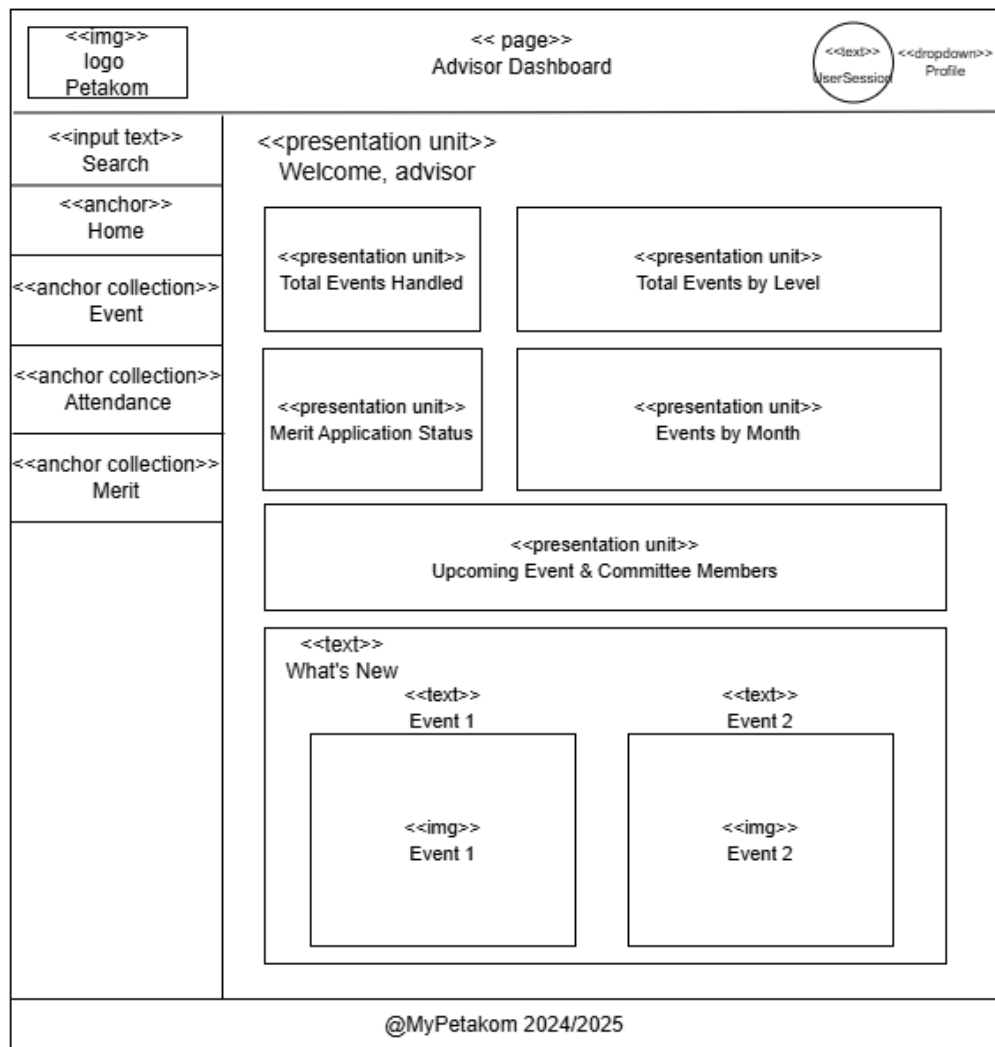


Figure 7.11 Petakom Advisor (Staff) Dashboard



Search

Home

Event

Attendance

Merit

New Event

UserSession

Profile

Event Registration

Event Title

Event Title

Date

Date

Search Location

Location

Find

Event Location

Geolocation

Event Level

Event Level

Event Description

Event Description

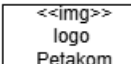
Upload Approval Letter

Approval Letter

Save Event

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Figure 7.12 New Event Page



Search

Home

Event

Attendance

Merit

View Event

UserSession

Profile

My Events

Search by Title

All Status

Search

Title	Status	Actions
event_title	event_status	View Details Delete

@MyPetakom 2024/2025

Figure 7.13 View Event Page

View Event Details

UserSession

Profile

Back

Home

Event

Attendance

Merit

Event Details

Event ID	event_id
Event Title	event_title
Status	event_status
Date	event_date
Geolocation	event_geolocation
Event Level	event_level
Event Description	event_description
Approval Letter	event_approval

Edit Event

Generate QR Code

@MyPetakom 2024/2025

Figure 7.14 View Event Details Page

View QR

UserSession

Profile

Home

Event

Attendance

Merit

QR Code for Event

Back to Event

@MyPetakom 2024/2025

Figure 7.15 View QR Page

Edit Event

UserSession

Profile

[Home](#)

[Event](#)

[Attendance](#)

[Merit](#)

Update Event

Event Title

Date

Search Location

Event Location

Event Level

Event Level

Event Description

Upload Approval Letter

Update

Cancel

@MyPetakom 2024/2025

Figure 7.16 Edit Event Page

Assign Committee

UserSession

Profile

[Home](#)

[Event](#)

[Attendance](#)

[Merit](#)

Assign Committee

Event	Date	Actions
event_title	event_date	View Edit Delete
event_title	event_date	Assign

@MyPetakom 2024/2025

Figure 7.18 Assign Committee Page



Search

Home

Event

Attendance

Merit

Back

Assign Committee for Event

Student ID

Position

Committee Role

Delete

Add Another Member

Add Committee

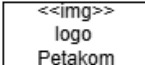
UserSession

Profile

Add Committee

@MyPetakom 2024/2025

Figure 7.19 Add Committee Page



Search

Home

Event

Attendance

Merit

Back

Committee Members for Event

Student Name	Student ID	Position
user_name	student_id	committee_role

Add Committee

UserSession

Profile

@MyPetakom 2024/2025

Figure 7.20 View Committee Page

Edit Committee

UserSession

Profile

Search

[Home](#)

[Event](#)

[Attendance](#)

[Merit](#)

Edit Committee for Event

Student ID

Position

Committee Role

Delete

Add Another Member

Save Changes

Cancel

@MyPetakom 2024/2025

Figure 7.21 Edit Committee Page

Apply Merit

UserSession

Profile

Search

[Home](#)

[Event](#)

[Attendance](#)

[Merit](#)

Apply Merit for My Events

Search by Title

All Status

Search

Event Title	Date	Status	Action
event_title	event_date	merit_status	View Details
event_title	event_date	merit_status	Apply

@MyPetakom 2024/2025

Figure 7.22 Apply Merit Page



page

Apply Merit Form

UserSession

Profile

Back

presentation unit

Apply Merit for Event

Event Details

Event ID	event_id
Title	event_title
Date	event_date
Geolocation	event_geolocation
Level	event_level
Event Description	event_description
Approval Letter	event_approval

Committee Members

Name	Student ID	Role	Event Level	Merit Score
user_name	student_id	committee_role	event_level	merit_score

Apply

Figure 7.23 Apply Merit Form Page

<< page>>
View Merit Application

<<presentation unit>>
Merit Application Details for Event

<<text>> Event Details

<<text>> Event ID	<<presentation unit>> event_id
<<text>> Title	<<presentation unit>> event_title
<<text>> Date	<<presentation unit>> event_date
<<text>> Geolocation	<<presentation unit>> event_geolocation
<<text>> Level	<<presentation unit>> event_level
<<text>> Event Description	<<presentation unit>> event_description
<<text>> Approval Letter	<<presentation unit>> event_approval

<<text>> Committee Members

<<text>> Name	<<text>> Student ID	<<text>> Role	<<text>> Event Level	<<text>> Merit Score
<<presentation unit>> user_name	<<presentation unit>> student_id	<<presentation unit>> committee_role	<<presentation unit>> event_level	<<presentation unit>> merit_score

@MyPetakom 2024/2025

Figure 7.24 View Merit Application Page

<>
logo
Petakom

<< page>>
Merit Application Admin

<<text>>
UserSession

<<dropdown>>
Profile

<<input text>>
Search

<<anchor>>
Home

<<anchor collection>>
Membership

<<anchor collection>>
Event

<<anchor collection>>
Merit

<<presentation unit>>
Merit Applications

<<text>> Event Title	<<text>> Event ID	<<text>> Event Date	<<text>> View Details	<<text>> Action
<<presentation unit>> event_title	<<presentation unit>> event_id	<<presentation unit>> event_date	<<button>> View Details	<<dropdown>> merit_status <<button>> Apply Changes

@MyPetakom 2024/2025

Figure 7.25 Merit Application Admin Page

 Student View Event UserSession Profile							
Search Home Event Merit	Joined Events Search by Title All Roles Search <table border="1"> <thead> <tr> <th>Event Title</th> <th>Role</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td>event_title</td> <td>committee_role</td> <td>View Details</td> </tr> </tbody> </table>	Event Title	Role	Action	event_title	committee_role	View Details
Event Title	Role	Action					
event_title	committee_role	View Details					
@MyPetakom 2024/2025							

Figure 7.26 Student View Event Page

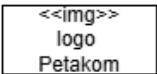
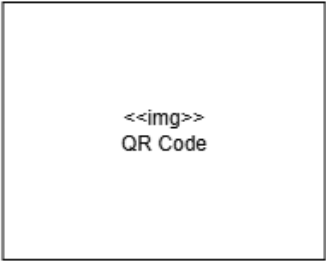
 Student Event Details UserSession Profile	
Search Home Event Attendance Merit	Back Event Details Scan the QR code to get the event details! 
@MyPetakom 2024/2025	

Figure 7.27 Student Event Details Page

<< page>> Student Event QR Display	
<<presentation unit>> Event Details	
<<text>> Title:	<<presentation unit>> event_title
<<text>> Date	<<presentation unit>> event_date
<<text>> Location	<<presentation unit>> event_location
<<text>> Description	<<presentation unit>> event_description

Figure 7.28 Student Event QR Display Page

7.3 Module 3 (Noora Haidah Binti Azmi – CB23095)

<> logo Petakom		<< page>> View Attendance		<<class>> UserSession	<<dropdown>> Profile
<<input text>> Search					<<button>> + Create
<<anchor>> Home	<<presentation unit>> Existing Attendance Slots				
<<anchor collection>> Event	<<text>> Event:	<<presentation unit>> event_title(event_id)			
<<anchor collection>> Attendance	<<presentation unit>> slot_time	<<button>> View Details	<<button>> Attendance	<<button>> Update	<<button>> Delete
<<anchor collection>> Merit					
@MyPetakom 2024/2025					

Figure 7.29 Petakom Advisor (Staff) View Attendance page

	<< page>> View Attendance	<<text>> UserSession <<dropdown>> Profile
<<input text>> Search	<<button>> Back <<presentation unit>> Create Attendance Slot <<text>> Select Event: <<input text>> Select Event <<text>> Date: <<input field>> Date <<text>> Time: <<input field>> Time <<text>> Search Location Name: <<input text>> Location <<button>> Search Location <<text>> Pick Location on Maps: <<input field>> Geolocation <<text>> Latitude: <<input text>> Latitude <<text>> Longitude: <<input text>> Longitude <<button>> Submit	
<<anchor>> Home		
<<anchor collection>> Event		
<<anchor collection>> Attendance		
<<anchor collection>> Merit		
@MyPetakom 2024/2025		

Figure 7.30 Petakom Advisor (Staff) Create Attendance Slot page

 View Details UserSession Profile	
Search Home Event Attendance Merit	Back Event Attendance Slot Details Event: event_title Date: slot_time Time: slot_time Geolocation: slot_geolocation QR Code Download QR Print QR
@MyPetakom 2024/2025	

Figure 7.31 Petakom Advisor (Staff) View Details

Attendance Slot List-MyPetakom

Slot ID:

attendanceslot_id

Event ID:

event_id

Event Title:

event_title

Current Malaysia Time:

checkin_time

Verify your attendance:

Please enter your student ID and password

Student ID:

Student Id

Password:

Password

Verify

Figure 7.32 Attendance Slot QR page

Attendance

UserSession

Profile

Back

Home

Event

Attendance

Merit

Change Attendance Location Verification Status

Student ID

Update All

Attendance ID	Slot ID	Event Title	Check-in Time	Location Verification
attendance_id	attendanceslot_id	event_title	checkin_time	location_verification

@MyPetakom 2024/2025

Figure 7.35 Petakom Advisor (Staff) Attendance page

localhost says

Are you sure want to delete this slot?

OK

Cancel

UserSession

Profile

+ Create

Home

Event

Attendance

Merit

Existing Attendance Slots

Event:

View Details

Attendance

Update

Delete

@MyPetakom 2024/2025

Figure 7.36 Petakom Advisor (Staff) Delete Existing Attendance Slot page

Attendance Records

UserSession

Profile

Search

[Home](#)

[Membership](#)

[Event](#)

[Merit](#)

Select Event:
 Select Event

Event Attendance Verification Summary

Event Title	Verified Students	Unverified Students
event_title	location_verification	location_verification

Graph

Graph

@MyPetakom 2024/2025

Figure 7.37 Administrator Attendance Records page

7.4 Module 4 (Noora Hanim Binti Azmi – CB23040)

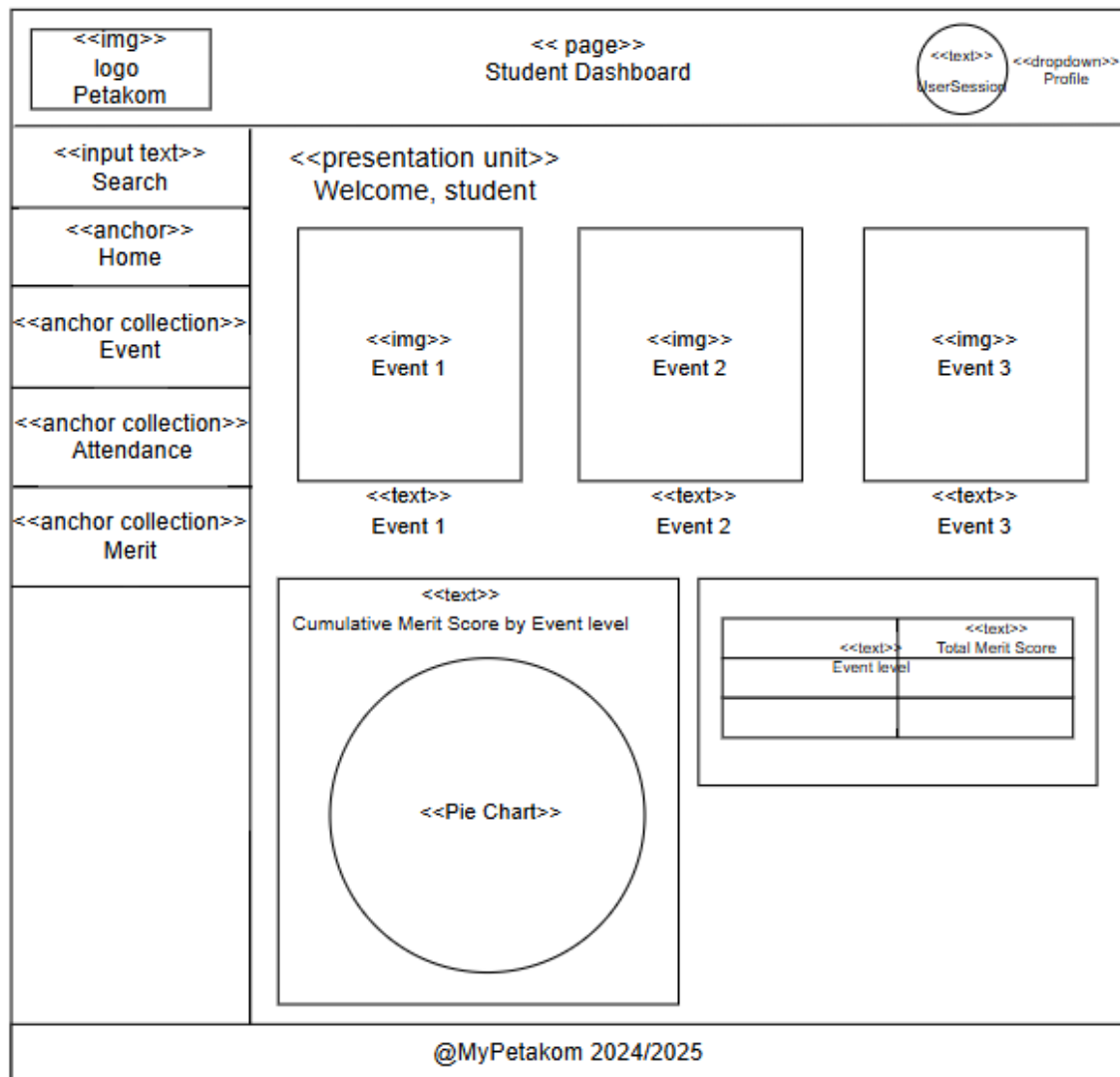


Figure 7.38 My Petakom Student Dashboard

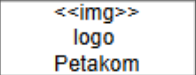
		<< page>> Missing Merit		UserSession	Profile
Search	Back	<<presentation unit>> MISSING MERIT APPLICATION		View Application	
Home	Student ID: Event Title: --Select Event-- Role: --Select Role-- Upload Supporting Document: --Select Event-- Submit				
Event					
Attendance					
Merit					
@MyPetakom 2024/2025					

Figure 7.39 Missing Merit Application page

		<< page>> View Missing Merit		UserSession	Profile																		
Search	Back	Search For Event Title	All	In Progress	Submitted	+Create																	
Home	<<presentation unit>> MISSING MERIT APPLICATION																						
Event	<table border="1"> <thead> <tr> <th>Student ID</th> <th>Claim ID</th> <th>Event ID</th> <th>Event Title</th> <th>Role</th> <th>Merit Score</th> <th>Claim Status</th> <th>Upload Letter</th> <th>Action</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> <td> Update Delete </td> </tr> </tbody> </table>					Student ID	Claim ID	Event ID	Event Title	Role	Merit Score	Claim Status	Upload Letter	Action									Update Delete
Student ID	Claim ID	Event ID	Event Title	Role	Merit Score	Claim Status	Upload Letter	Action															
								Update Delete															
Attendance																							
Merit																							
@MyPetakom 2024/2025																							

Figure 7.40 Display Missing Merit Application page

Missing Merit

UserSession

Profile

Back

UPDATE MISSING MERIT

View Application

Home

Event

Attendance

Merit

Student ID:

Event Title:

--Select Event--

Role:

--Select Role--

Upload Supporting Document:

Event Letter

Submit

@MyPetakom 2024/2025

Figure 7.41 Update Missing Merit Application page

Delete this merit claim?

OK

Cancel

UserSession

Profile

Back

All

In Progress

Submitted

+Create

Home

Event

Attendance

Merit

MISSING MERIT APPLICATION

Student ID	Claim ID	Event ID	Event Title	Role	Merit Score	Claim Status	Upload Letter	Action
								Update Delete

@MyPetakom 2024/2025

Figure 7.42 Delete Missing Merit Application page

<>
 logo
 Petakom

<<presentation unit>>
 Student Merit Details

<<text>> Name:	<<Text>>>
<<text>> Student ID:	<<Text>>>
<<text>> Programme:	<<Text>>>
<<text>> Total Merits:	<<Text>>>

<<text>> Event Merit Breakdown

<<text>> Event Title:	<<text>> Event Date:	<<text>> Role:	<<text>> Merit Score:
<<Text>>>	<<Text>>>	<<Text>>>	<<Text>>>
<<Text>>>	<<Text>>>	<<Text>>>	<<Text>>>
<<Text>>>	<<Text>>>	<<Text>>>	<<Text>>>

<<button>>
 Print as PDF

Figure 7.45 Display the Student Merit Details page After Scan QR Code



Update Merit Claim Status

UserSession

Profile

Back

Home

Event

Attendance

Merit

Change Merit Claim Status

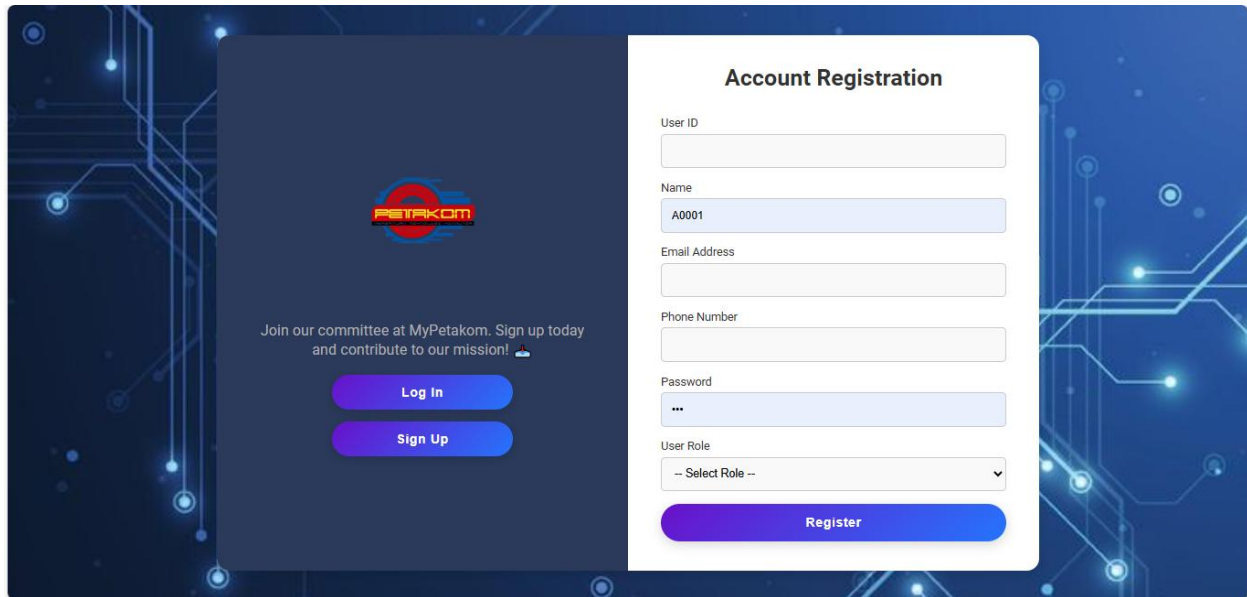
Claim ID ID	Student ID	Event Title	Letter	Current Status	Change Status	
					Status	Update
					Status	Update
					Status	Update
					Status	Update

@MyPetakom 2024/2025

Figure 7.46 Petakom Advisor (Staff) Update Merit Claim Status page.

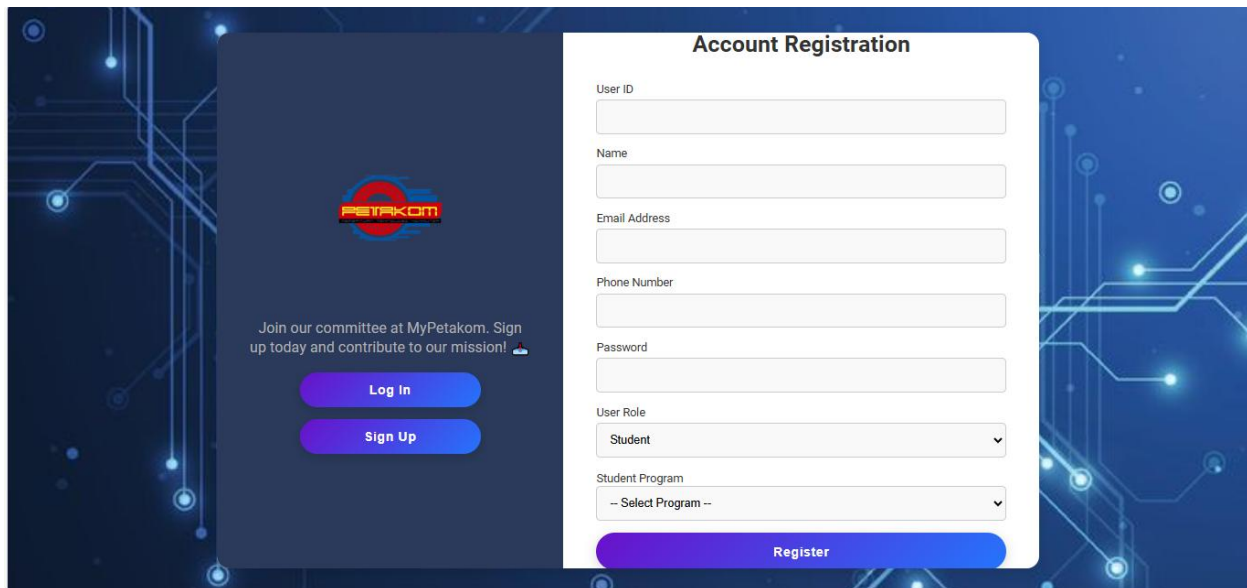
8.0 INTERFACE DESIGN

8.1 MODULE 1 (Siti Nurizzati Binti Zabidi – CA23075)



The image shows a web interface for account registration. On the left, a dark blue panel features the 'PETAKOM' logo and a message: 'Join our committee at MyPetakom. Sign up today and contribute to our mission!'. Below this are 'Log In' and 'Sign Up' buttons. On the right, a white panel titled 'Account Registration' contains the following fields: 'User ID' (text input), 'Name' (text input with 'A0001' entered), 'Email Address' (text input), 'Phone Number' (text input), 'Password' (text input with three dots), 'User Role' (dropdown menu with '-- Select Role --'), and a 'Register' button at the bottom.

Figure 8.1 Account Registration for Staff Interface



The image shows a web interface for account registration, similar to the one above but for students. The left panel is identical. The right panel, titled 'Account Registration', includes the same fields as the staff interface but with additional ones: 'Student Program' (dropdown menu with '-- Select Program --') and a 'Register' button at the bottom. The 'User Role' dropdown is set to 'Student'.

Figure 4.2 Account Registration for Student Interface

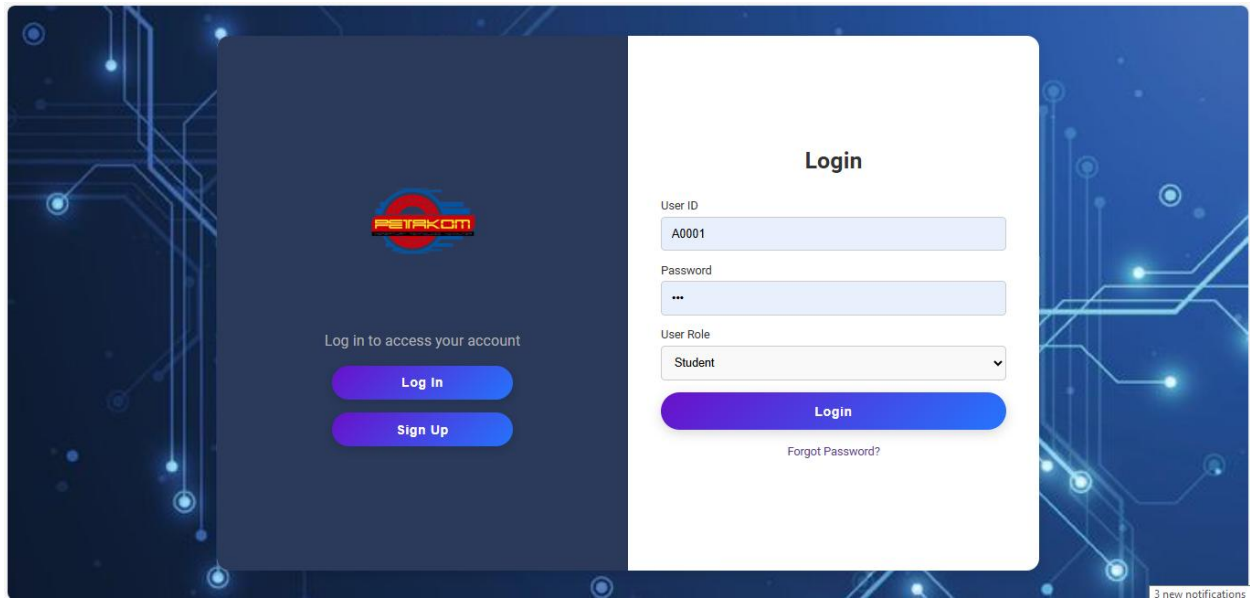


Figure 8.3 Login Page for users Interface

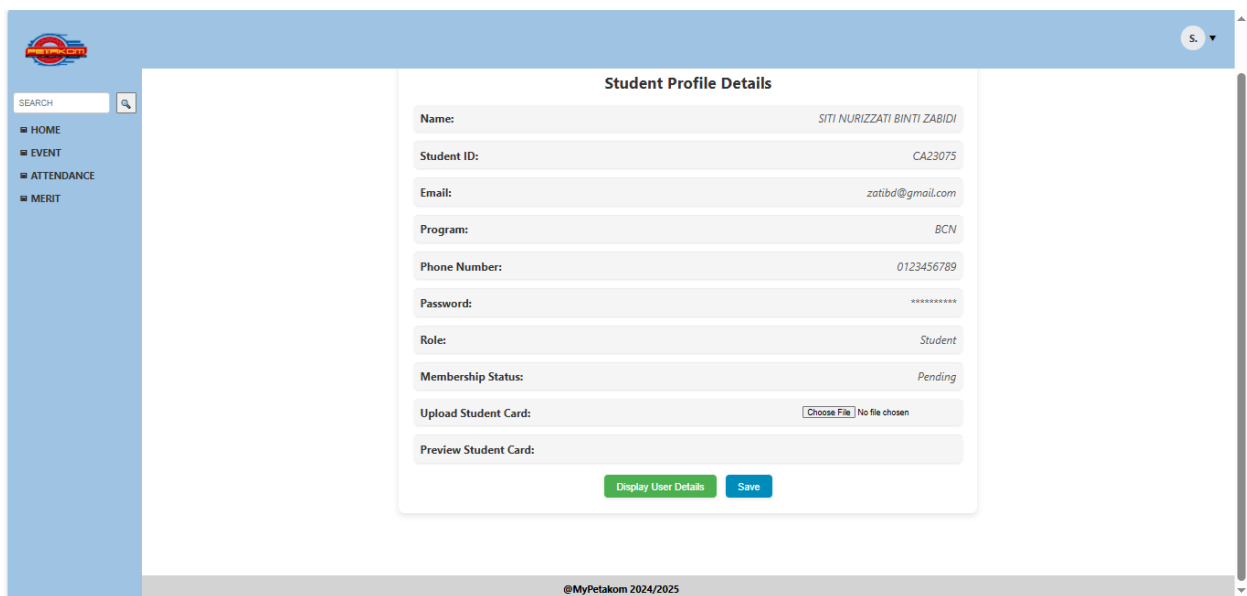


Figure 8.4 Student's Profile Details Interface

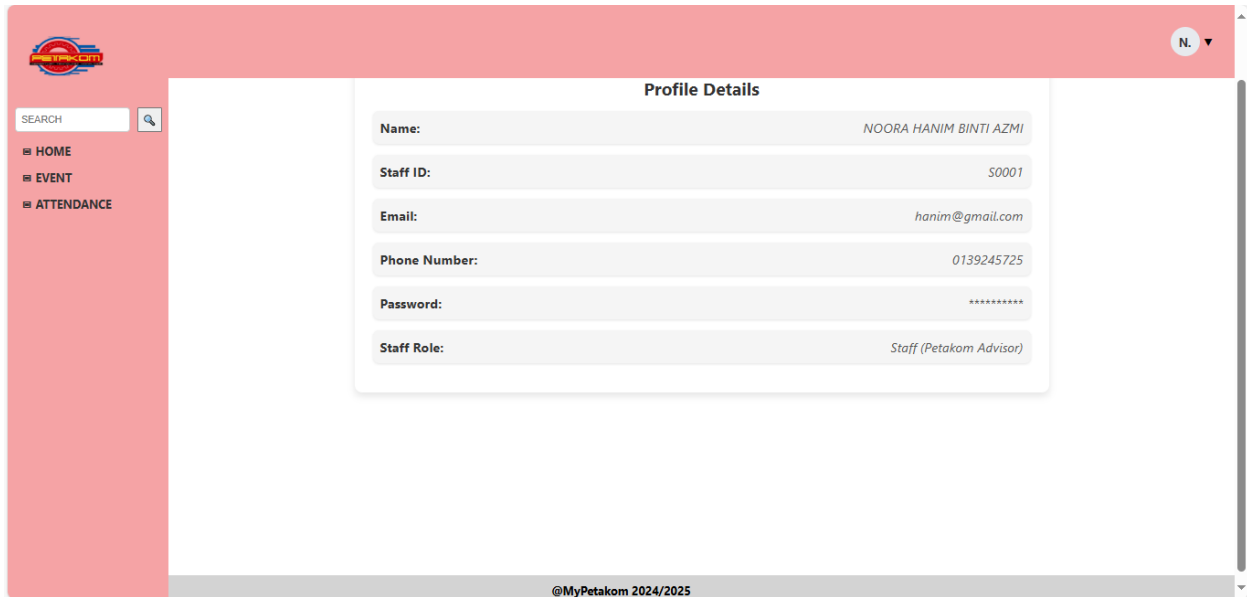


Figure 8.5 Staff (Petakom Advisor)'s Profile Details Interface

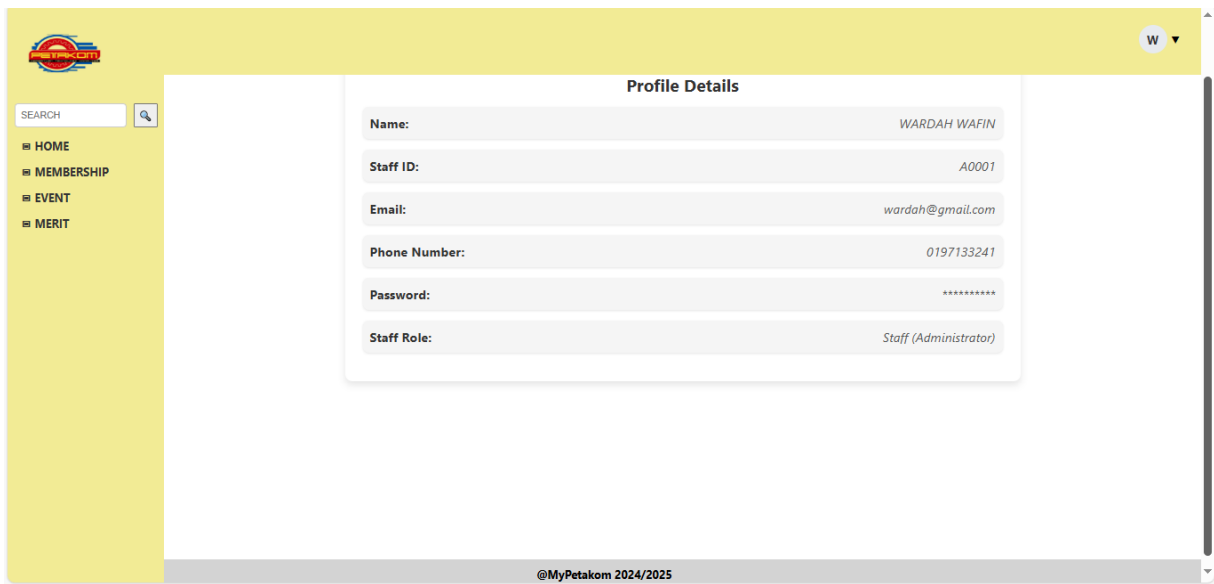


Figure 8.6 Staff (Administrator)'s Profile Details Interface

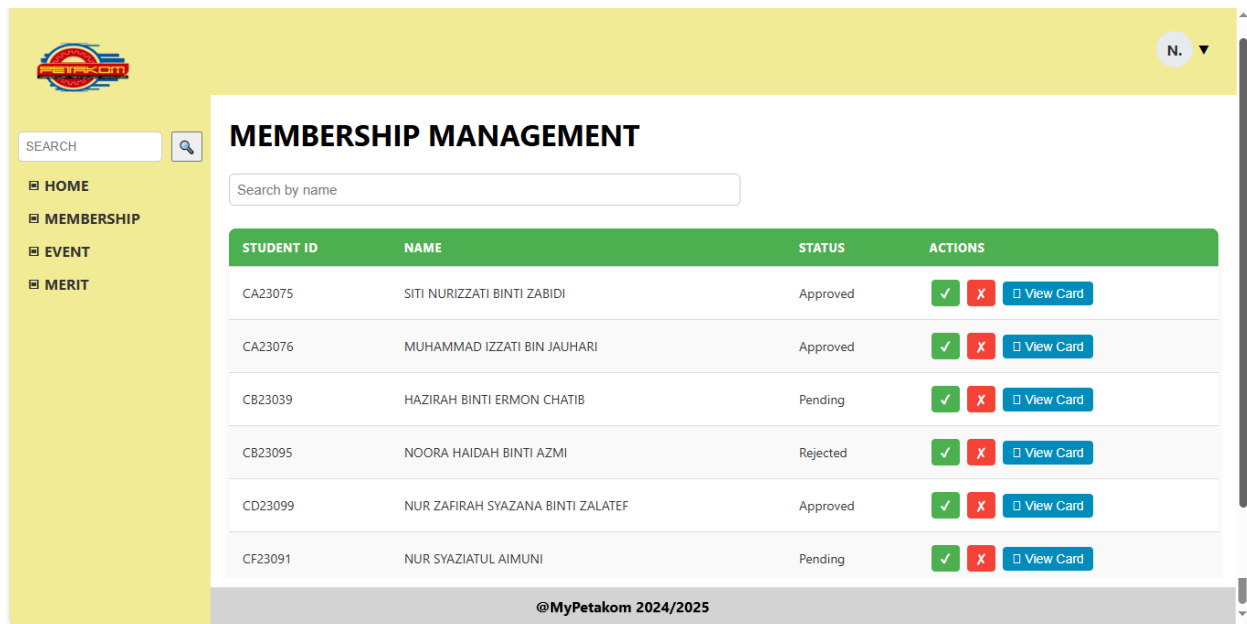


Figure 8.7 Membership Verification Interface

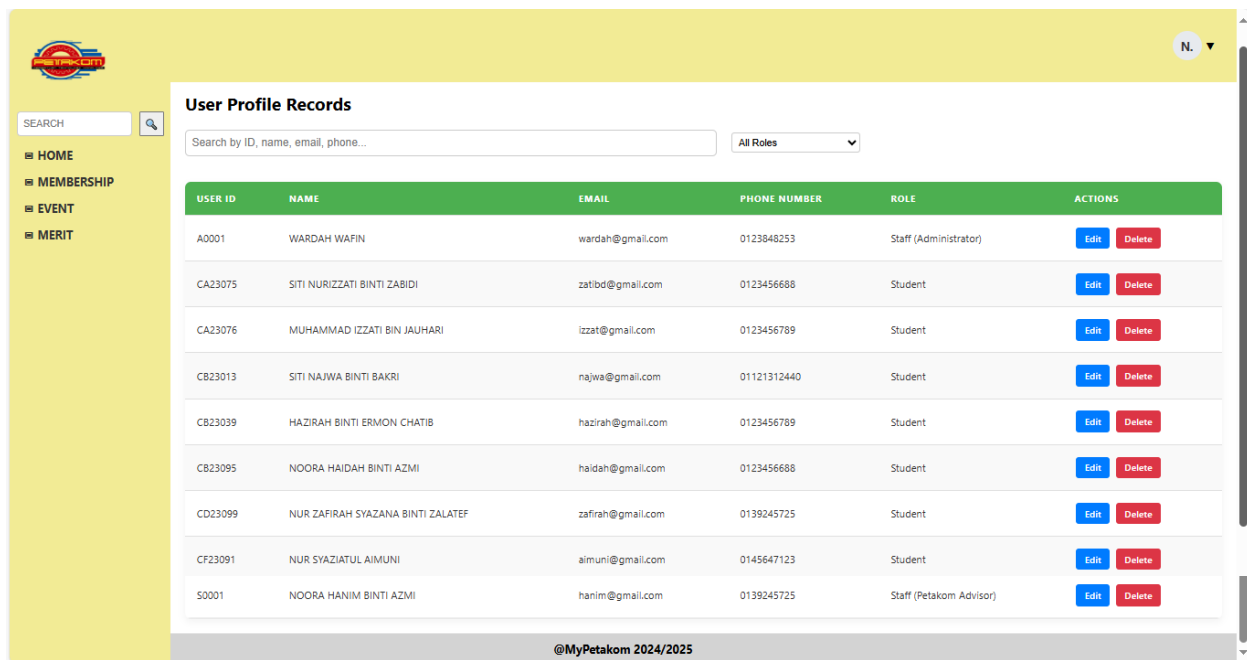


Figure 8.8 List of user Interface

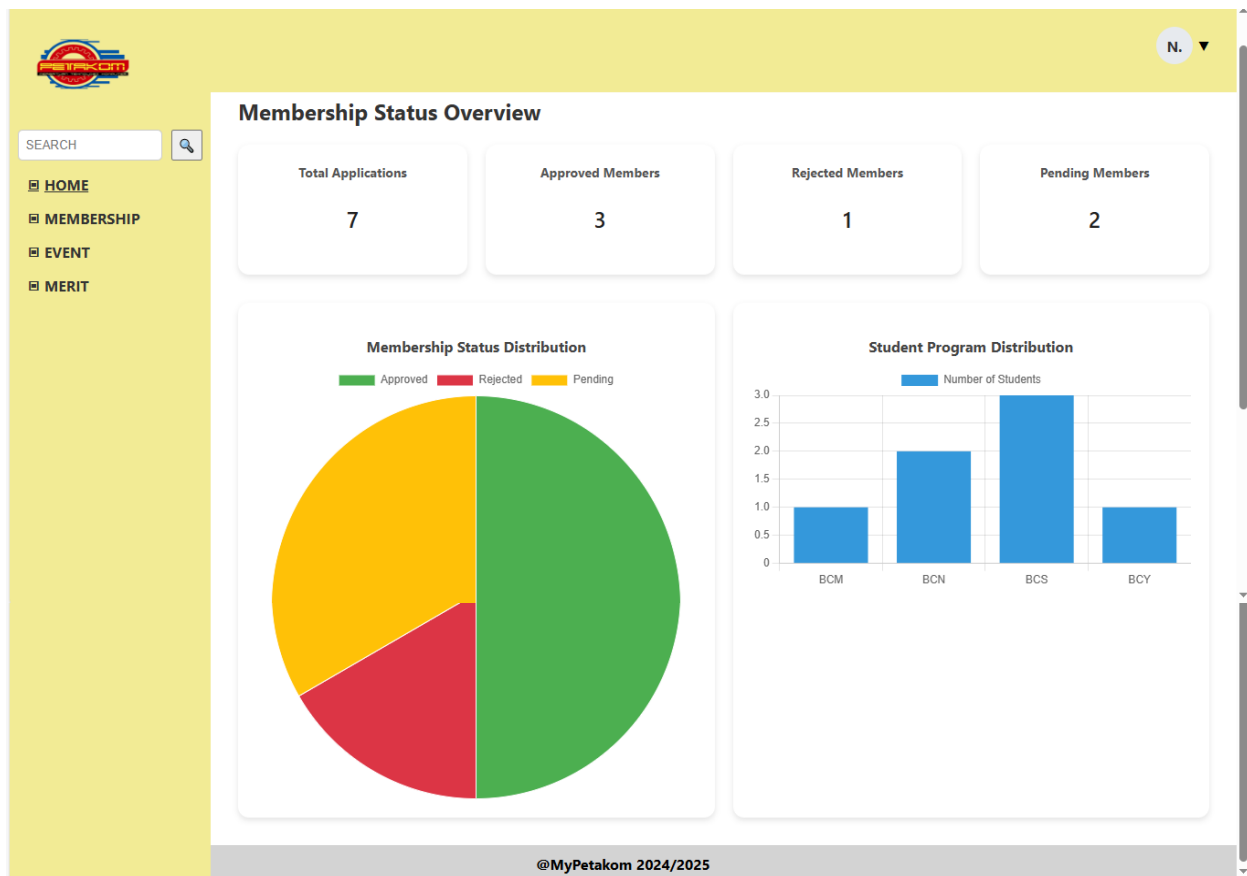


Figure 8.9 Staff (Administrator) Dashboard Interface

The "Edit User Profile" form includes input fields for Email (wardah@gmail.com) and Phone Number (0123848253), along with Save and Cancel buttons.

@MyPetakom 2024/2025

Figure 8.10 Edit User Profile Interface

8.2 MODULE 2 (Siti Najwa Binti Bakri – CB23013)

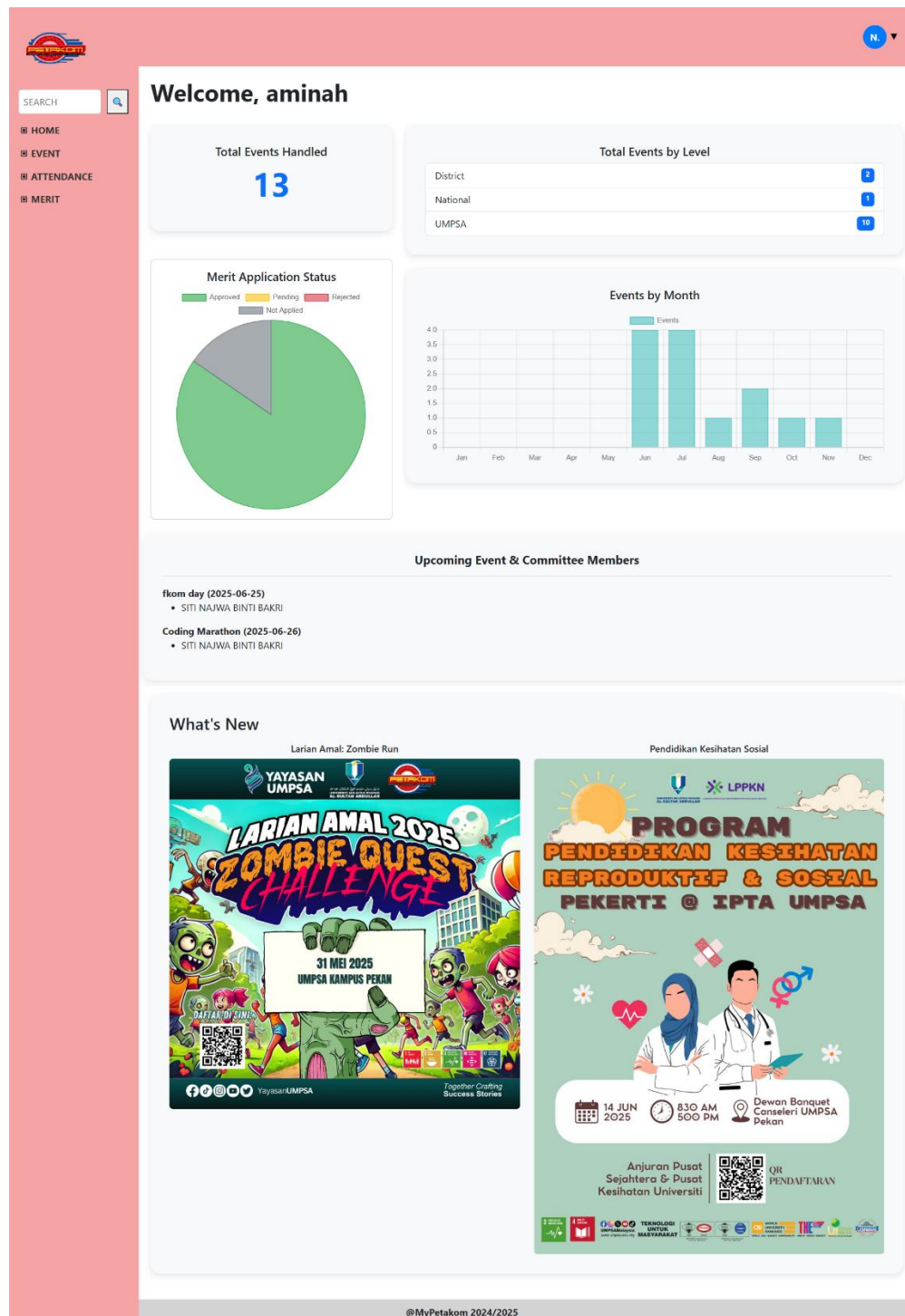
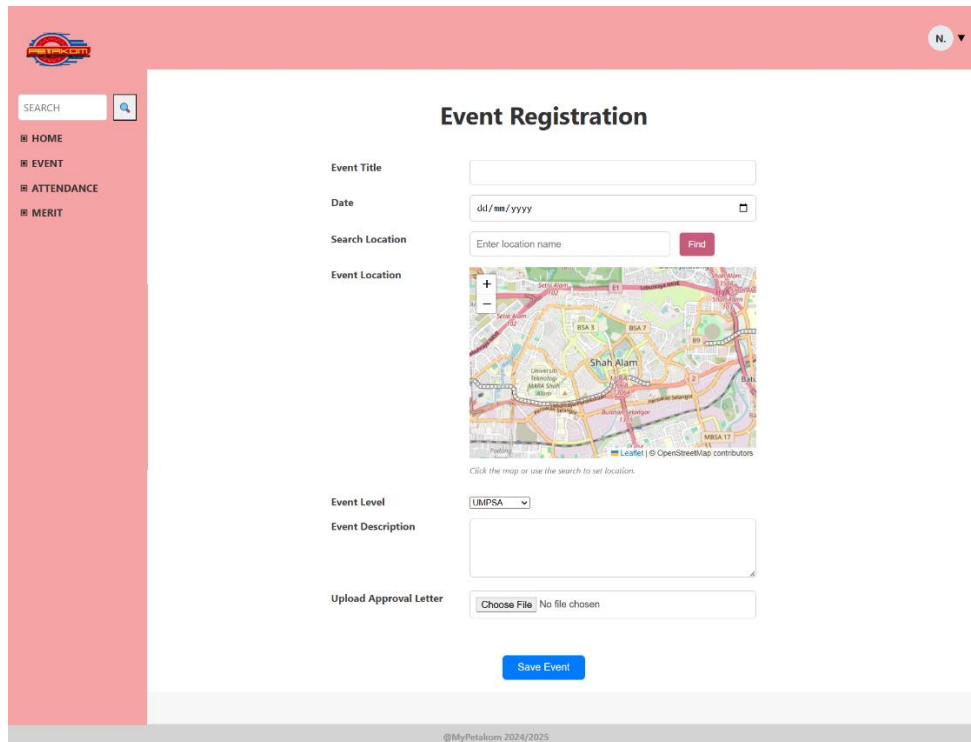


Figure 8.11 Petakom Advisor (Staff) Dashboard Interface

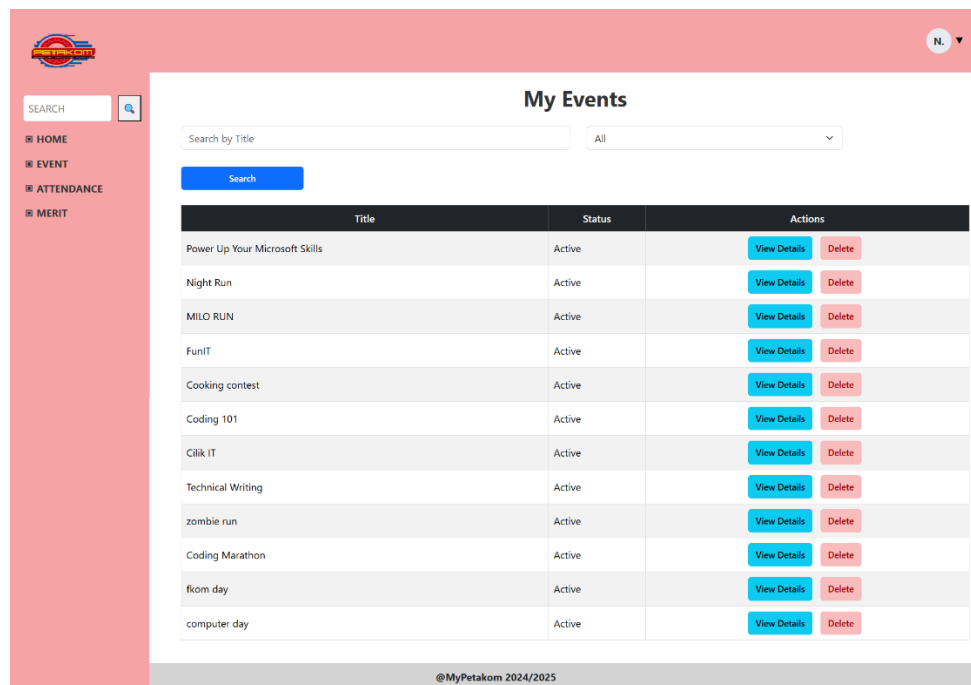


The Event Registration interface features a sidebar with navigation links: HOME, EVENT, ATTENDANCE, and MERIT. The main content area is titled "Event Registration" and contains the following fields:

- Event Title:** A text input field.
- Date:** A date picker showing "dd/mm/yyyy".
- Search Location:** A text input field with a "Find" button.
- Event Location:** A map showing the Shah Alam area with a "Click the map or use the search to set location." instruction.
- Event Level:** A dropdown menu currently set to "UMPSA".
- Event Description:** A large text area.
- Upload Approval Letter:** A "Choose File" button and "No file chosen" text.

A "Save Event" button is located at the bottom of the form. The footer indicates "@MyPetakom 2024/2025".

Figure 8.12 Event Registration Interface



The View Event interface features a sidebar with navigation links: HOME, EVENT, ATTENDANCE, and MERIT. The main content area is titled "My Events" and includes a search bar and a table of events.

Search by Title: All

Title	Status	Actions
Power Up Your Microsoft Skills	Active	View Details Delete
Night Run	Active	View Details Delete
MILO RUN	Active	View Details Delete
FunIT	Active	View Details Delete
Cooking contest	Active	View Details Delete
Coding 101	Active	View Details Delete
Clak IT	Active	View Details Delete
Technical Writing	Active	View Details Delete
zombie run	Active	View Details Delete
Coding Marathon	Active	View Details Delete
fkom day	Active	View Details Delete
computer day	Active	View Details Delete

The footer indicates "@MyPetakom 2024/2025".

Figure 8.13 View Event Interface

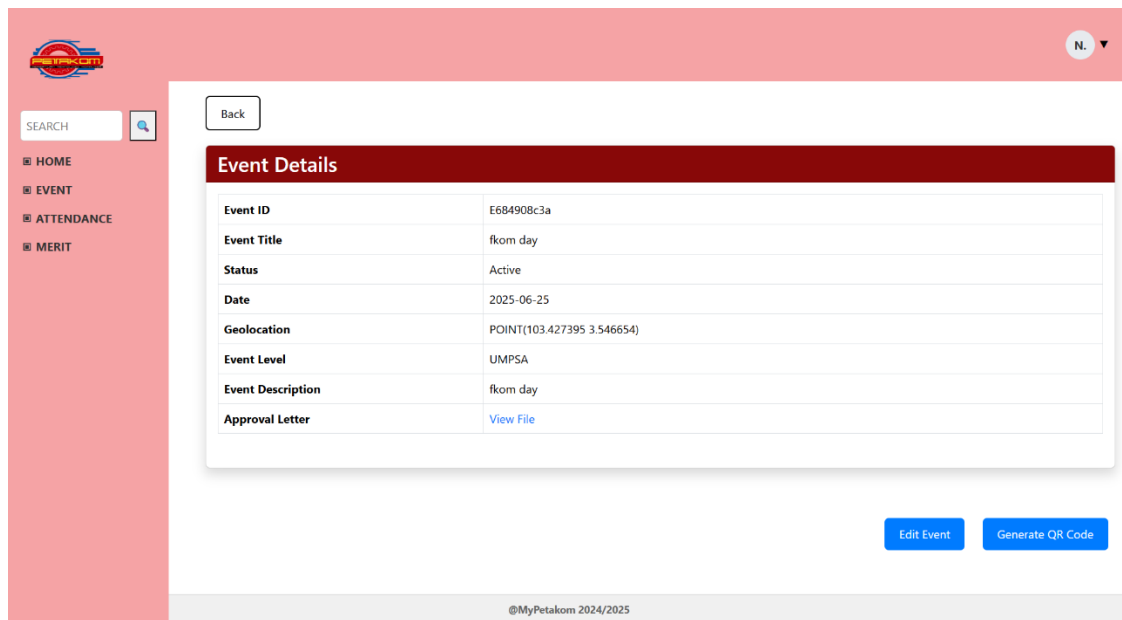


Figure 8.14 View Event Details Interface

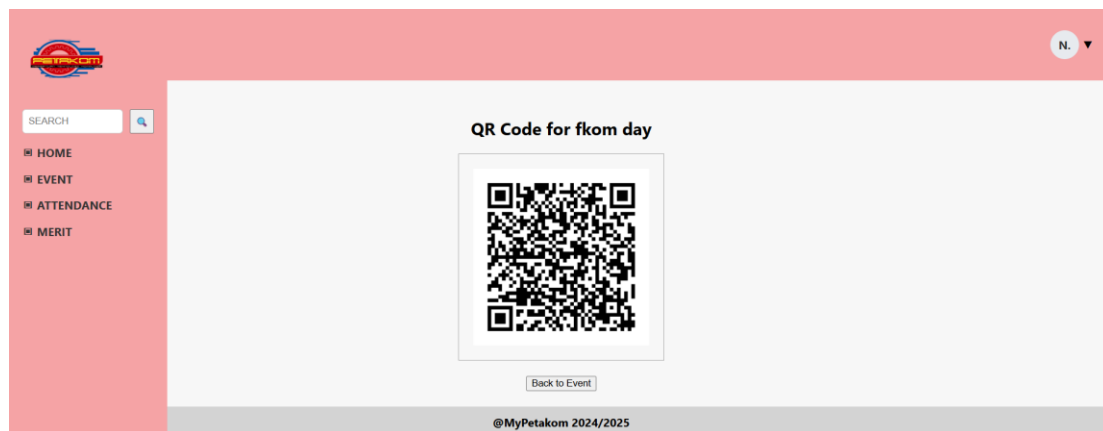


Figure 8.15 View QR Interface



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

+

-



POINT(103.427395 3.546654)

Event Title

fkorn day

Date

25/06/2025

Search Location

Enter location name

Find

Geolocation

POINT(103.427395 3.546654)

Event Location



Event Description

fkorn day

Upload Approval Letter

Current File: [View PDF](#)

☐ Remove current file

Update

Cancel

@MyPetakom 2024/2025

Figure 8.16 Edit Event Interface



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

+

-



POINT(103.427395 3.546654)

Event

Date

Actions

Power Up Your Microsoft Skills	2025-11-06	View Edit Delete
Night Run	2025-10-13	View Edit Delete
MILO RUN	2025-09-21	View Edit Delete
FunIT	2025-09-17	View Edit Delete
Cooking contest	2025-08-16	Assign
Coding 101	2025-07-30	Assign
Cilik IT	2025-07-22	Assign
Technical Writing	2025-07-04	View Edit Delete
zombie run	2025-06-28	View Edit Delete
Coding Marathon	2025-06-26	View Edit Delete
fkorn day	2025-06-25	View Edit Delete
computer day	2025-06-19	View Edit Delete

@MyPetakom 2024/2025

Figure 8.17 Assign Committee Interface



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

Back

Assign Committee for Cilik IT

Student ID:

Position:

Main Committee

Delete

Add Another Member

Add Committee

@MyPetakom 2024/2025

Figure 8.18 Add Committee Interface



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

Back

Committee Members for Cilik IT

Student Name	Student ID	Position
SITI NAJWA BINTI BAKRI	CB23013	Main Committee
IZZATI	CA23075	Main Committee
NOORA HANIM BINTI AZMI	CB23040	Main Committee

Figure 8.19 View Committee Interface


N.

- HOME
- EVENT
- ATTENDANCE
- MERIT

Edit Committee for Cilik IT

Student ID:

SITI NAJWA BINTI BAKRI

Position:

Main Committee

Delete

Student ID:

IZZATI

Position:

Main Committee

Delete

Student ID:

NOORA HANIM BINTI AZMI

Position:

Main Committee

Delete

Add Another Member

Save Changes

Cancel

@MyPetakom 2024/2025

Figure 8.20 Edit Committee Interface


N.

- HOME
- EVENT
- ATTENDANCE
- MERIT

Apply Merit for My Events

All

Search

Event Title	Date	Status	Action
Power Up Your Microsoft Skills	2025-11-06	Approved	View Details
Night Run	2025-10-13	Approved	View Details
MILO RUN	2025-09-21	Approved	View Details
FunIT	2025-09-17	Approved	View Details
Cooking contest	2025-08-16	Approved	View Details
Coding 101	2025-07-30	Not Applied	Apply
Cilik IT	2025-07-22	Not Applied	Apply
Technical Writing	2025-07-04	Approved	View Details
zombie run	2025-06-28	Approved	View Details
Coding Marathon	2025-06-26	Approved	View Details
fkom day	2025-06-25	Approved	View Details
computer day	2025-06-19	Approved	View Details

@MyPetakom 2024/2025

Figure 8.21 Apply Merit Interface



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

Back

Apply Merit for Cilik IT

Event Details

Event ID	E6848f21e8
Title	Cilik IT
Date	2025-07-22
Geolocation	POINT(103.398158 4.786532)
Level	District
Description	Teaching primary school students coding
Approval Letter	Download Approval Letter (PDF)

Committee Members

Name	Student ID	Role	Event Level	Merit Score
SITI NAJWA BINTI BAKRI	CB23013	Main Committee	District	40
IZZATI	CA23075	Main Committee	District	40
NOORA HANIM BINTI AZMI	CB23040	Main Committee	District	40

Apply

@MyPetakom 2024/2025

Figure 8.22 Apply Merit Form Interface

Merit Application Details for Cilik IT

Event Details

Event ID	E6848f21e8
Title	Cilik IT
Date	2025-07-22
Geolocation	POINT(103.398158 4.786532)
Level	District
Description	Teaching primary school students coding
Merit Status	Pending
Approval Letter	View File

Committee Members

Name	Student ID	Role	Event Level	Merit Score
SITI NAJWA BINTI BAKRI	CB23013	Main Committee	District	40
IZZATI	CA23075	Main Committee	District	40
NOORA HANIM BINTI AZMI	CB23040	Main Committee	District	40

@MyPetakom 2024/2025

Figure 8.23 View Merit Application Interface

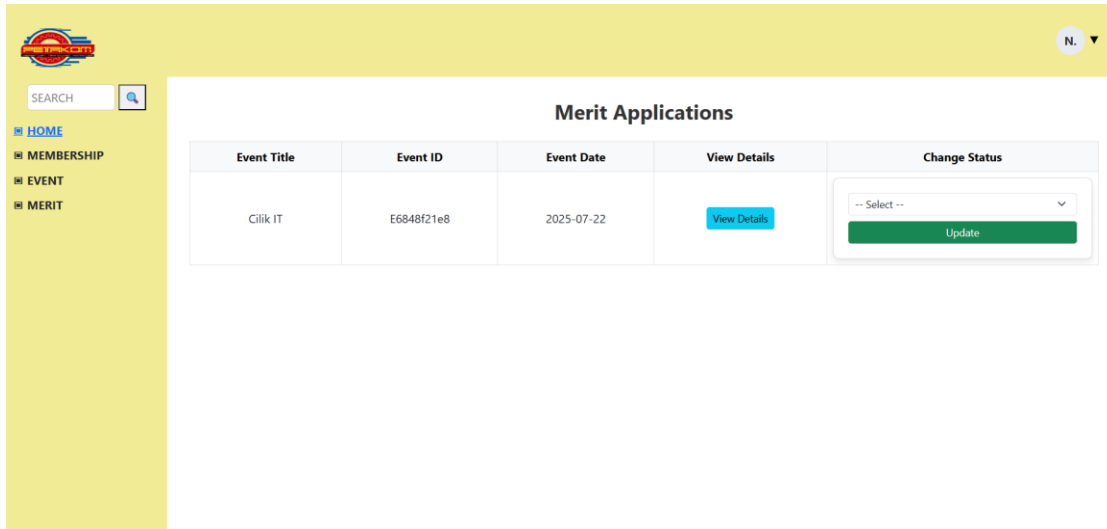


Figure 8.24 Merit Application Admin Interface

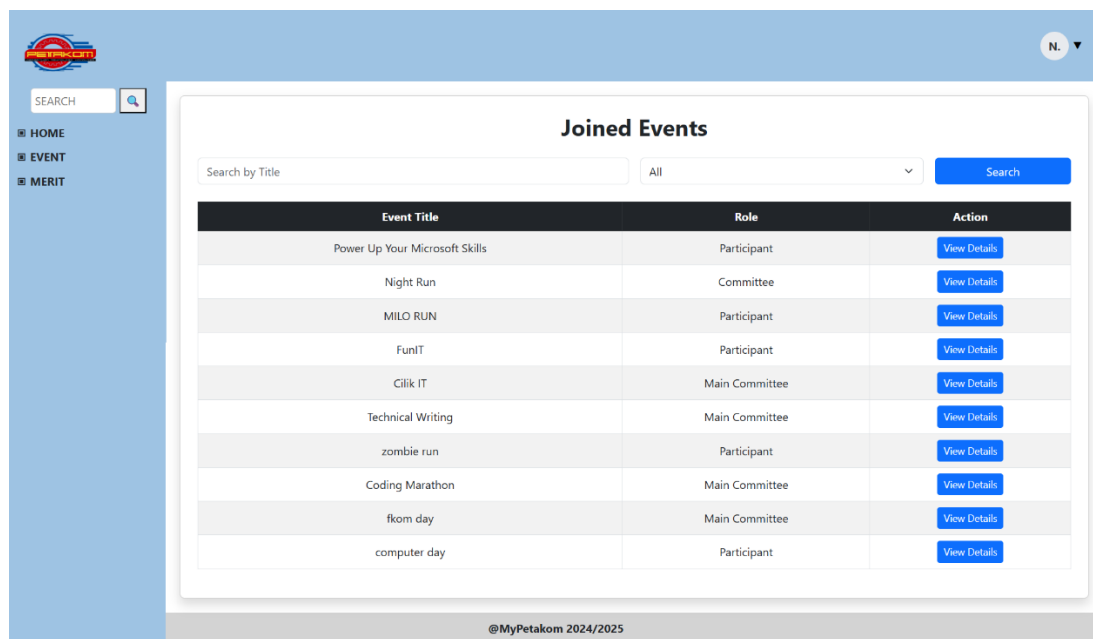


Figure 8.25 Student View Event Interface



Figure 8.26 Student Event Details Interface

Technical Writing Details	
Title:	Technical Writing
Date:	2025-07-04
Location:	Fakulti Komputeran UMPISA, Universiti Malaysia Pahang, Jalan Kuala Pahang - Tanah Putih, Kampung Marhum, Pekan, Pahang, 26600, Malaysia مليسيا
Description:	Teaching technical writing

Figure 8.27 Student Event QR Display Interface

8.3 MODULE 3 (Noora Haidah Binti Azmi – CB23095)

The screenshot displays the Petakom Advisor (Staff) View Attendance page. The interface features a red sidebar on the left with a search bar and navigation links: HOME, EVENT, ATTENDANCE, and MERIT. The main content area is titled 'Existing Attendance Slots' and shows two events. Each event has a table of attendance slots with columns for ID, Date, and Time. Action buttons (View Details, Attendance, Update, Delete) are provided for each slot. A '+ Create' button is located in the top right corner of the main content area. The footer of the page reads '@MyPetakom 2024/2025'.

Existing Attendance Slots		
Event: Night Run (ID: E6846ab7b7)		
1.	13 Oct 2025, 00:39	View Details Attendance Update Delete
Event: Power Up Your Microsoft Skills (ID: E68478f552)		
1.	06 Nov 2025, 16:54	View Details Attendance Update Delete

Figure 8.28 Petakom Advisor (Staff) View Attendance page



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

Back

Create Attendance Slot

Select Event:

-- Select Event --

Date:

dd/mm/yyyy

Time:

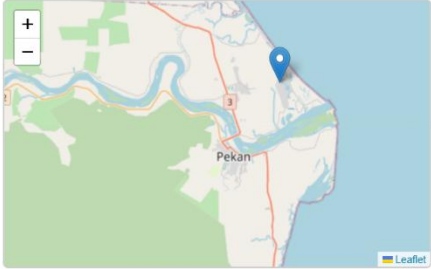
--:--:--

Search Location Name:

e.g. Universiti Malaysia Pahang

Search Location

Pick Location on Map:



Latitude:

3.546654

Longitude:

103.427395

Submit

Figure 8.29 Petakom Advisor (Staff) Create Attendance Slot page

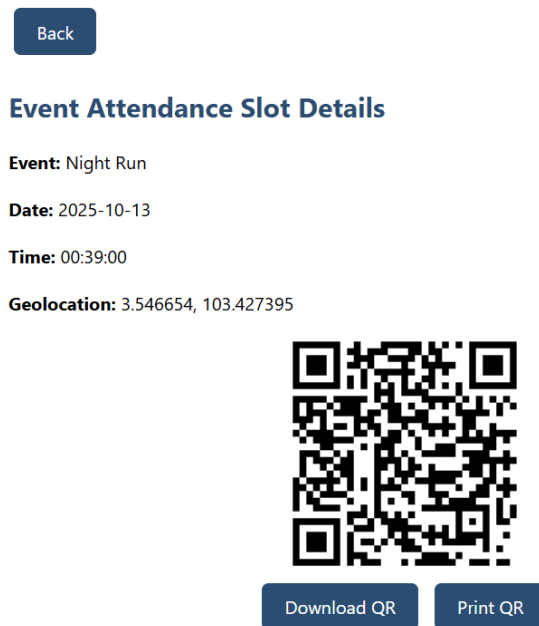


Figure 8.30 Petakom Advisor (Staff) View Details

Attendance Slot List - MyPetakom

Slot ID: AS0002

Event ID: E68468c2de

Event Title: Coding Marathon

Current Malaysia Time: 26/06/2025, 16:45:47

Verify your attendance:

Please enter your student ID and password

Student ID:

Password:

VERIFY

Figure 8.31 Attendance Slot QR page

Attendance Slot List - MyPetakom

Slot ID: AS0001

Event ID: E6846ab7b7

Event Title: Night Run

Current Malaysia Time: 26/06/2025, 16:54:39

Verify your attendance:

You have submitted your attendance!

Figure 8.32 Submit Attendance Slot QR page



SEARCH

Back

HOME

EVENT

ATTENDANCE

MERIT

SEARCH

Back

HOME

EVENT

ATTENDANCE

MERIT

Update Attendance Slot

Event ID:

E0846ab7b7

Date:

13/10/2025

Time:

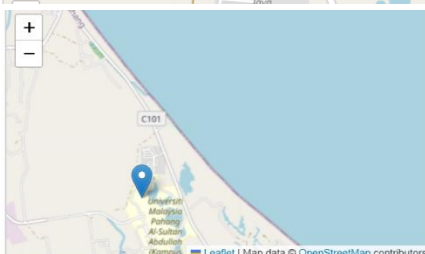
12:39 AM

Search Location Name:

e.g. Universiti Malaysia Pahang

Search Location

Pick Location on Map:



Latitude:

3.546654

Longitude:

103.427395

Update

@MyPetakom 2024/2025

Figure 8.33 Petakom Advisor (Staff) Update Attendance Slot page

Back

Change Attendance Location Verification Status

Student ID: CA23075 --Select Verification-- Update All				
Attendance ID	Slot ID	Event Title	Check-in Time	Location Verification
A001	AS0001	Night Run	2025-06-26 16:57:01	Verified

Student ID: CA23098 --Select Verification-- Update All				
Attendance ID	Slot ID	Event Title	Check-in Time	Location Verification
A002	AS0001	Night Run	2025-06-26 16:57:11	Not Verified

Student ID: CB23039 --Select Verification-- Update All				
Attendance ID	Slot ID	Event Title	Check-in Time	Location Verification
A003	AS0001	Night Run	2025-06-26 16:57:16	Verified

Figure 8.34 Petakom Advisor (Staff) Attendance page

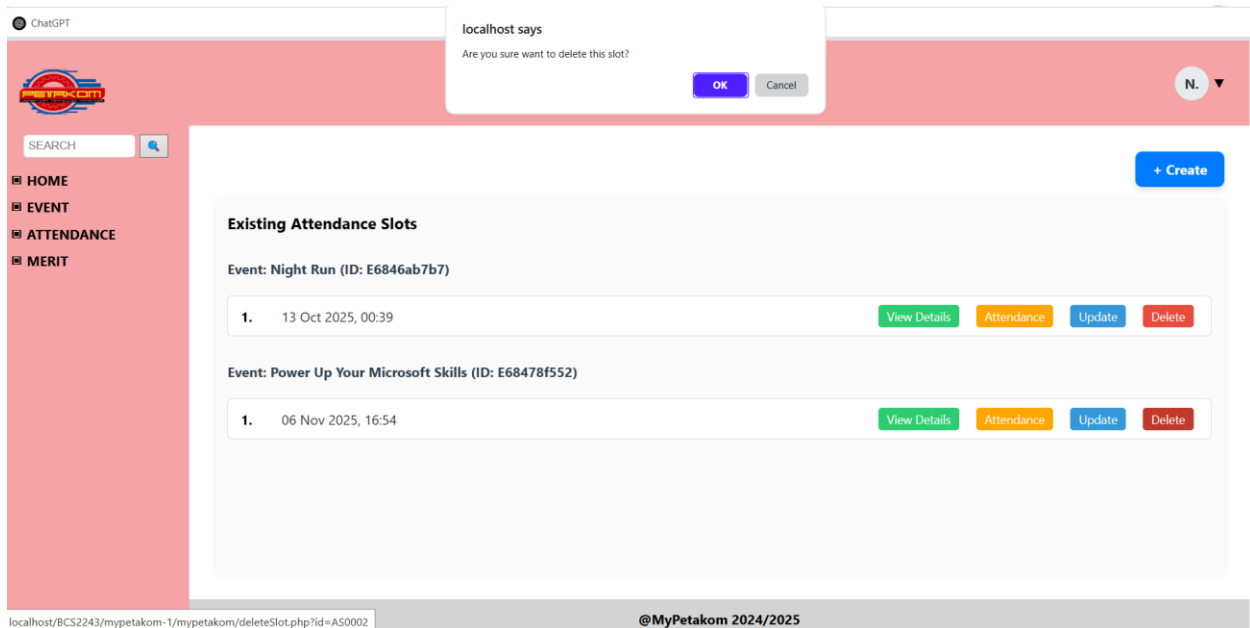


Figure 8.35 Petakom Advisor (Staff) Delete Existing Attendance Slot page

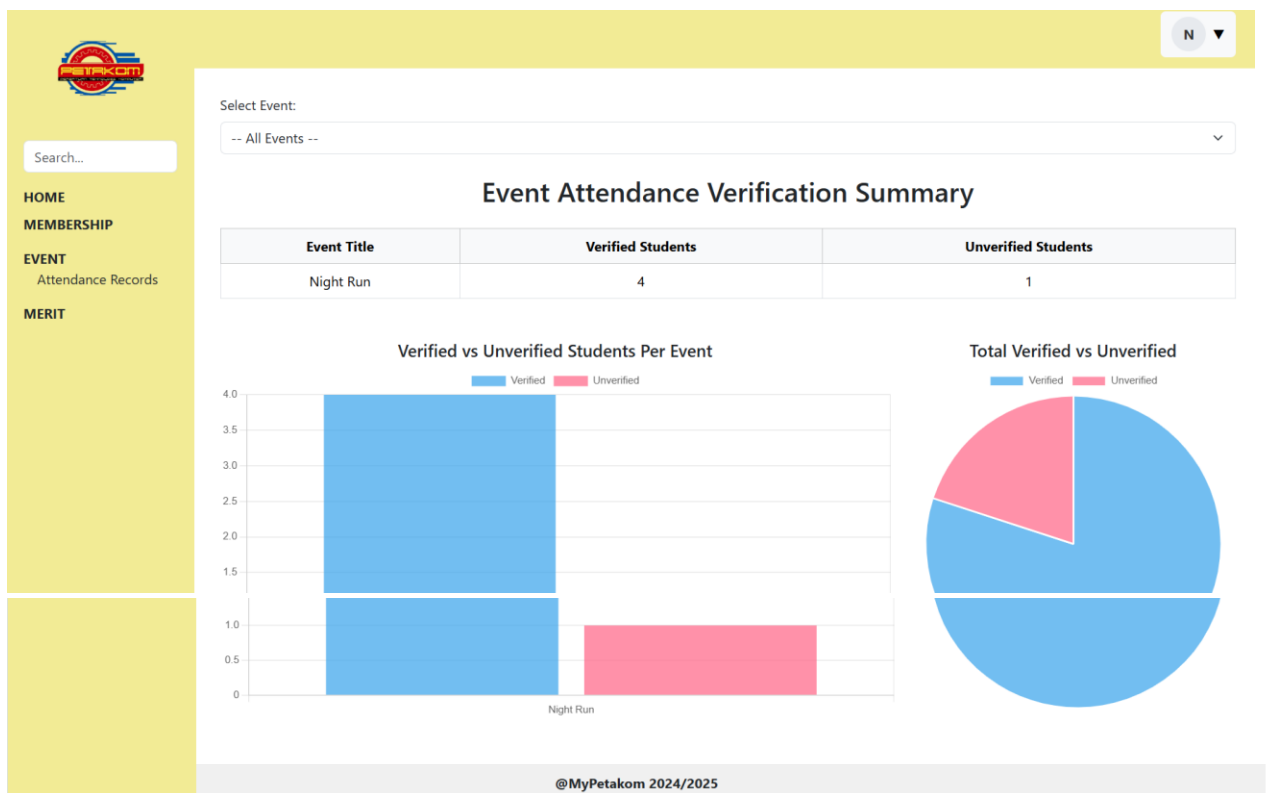


Figure 8.36 Administrator Attendance Records page

8.4 MODULE 4 (Noora Hanim Binti Azmi – CB23040)

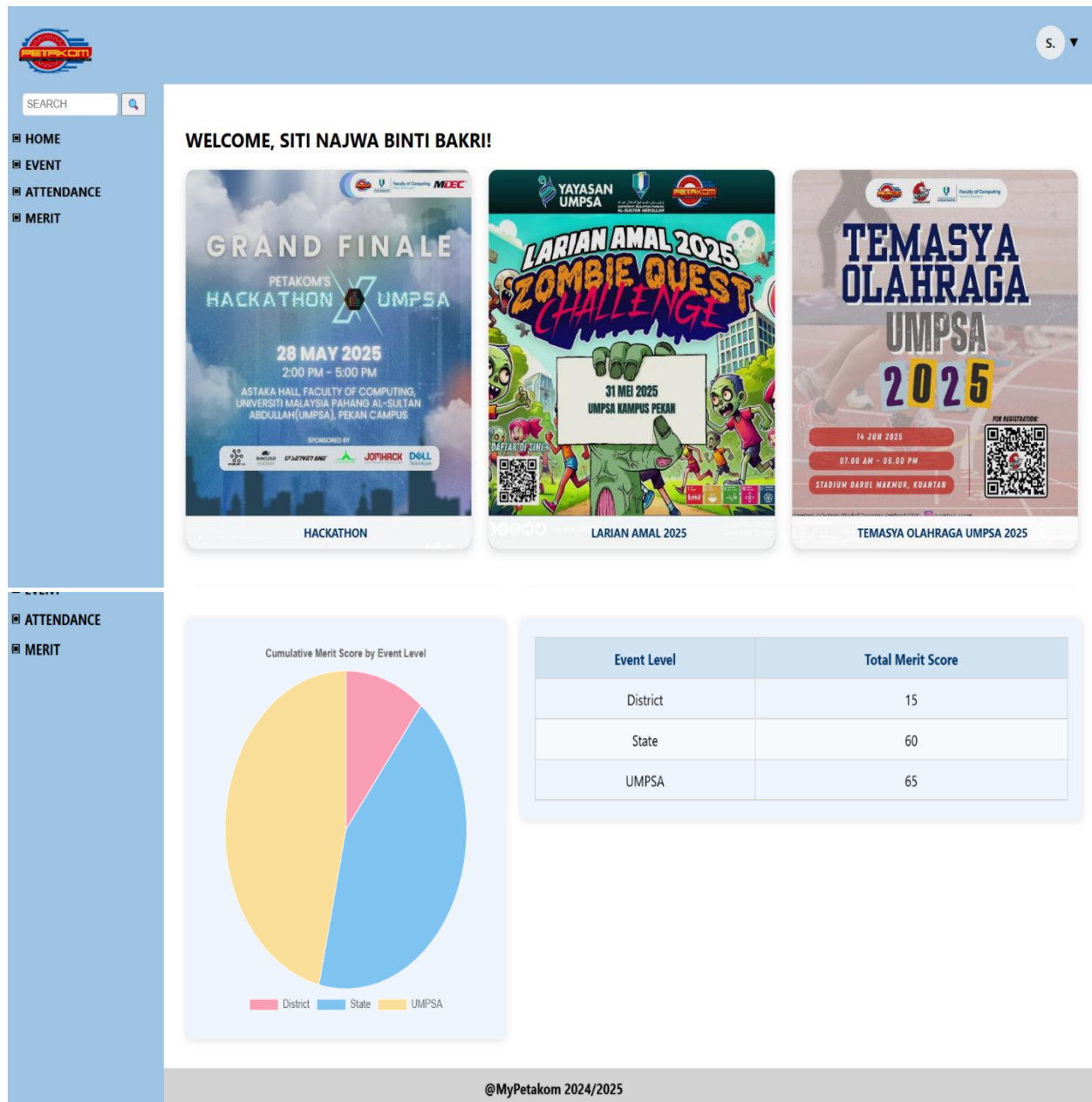
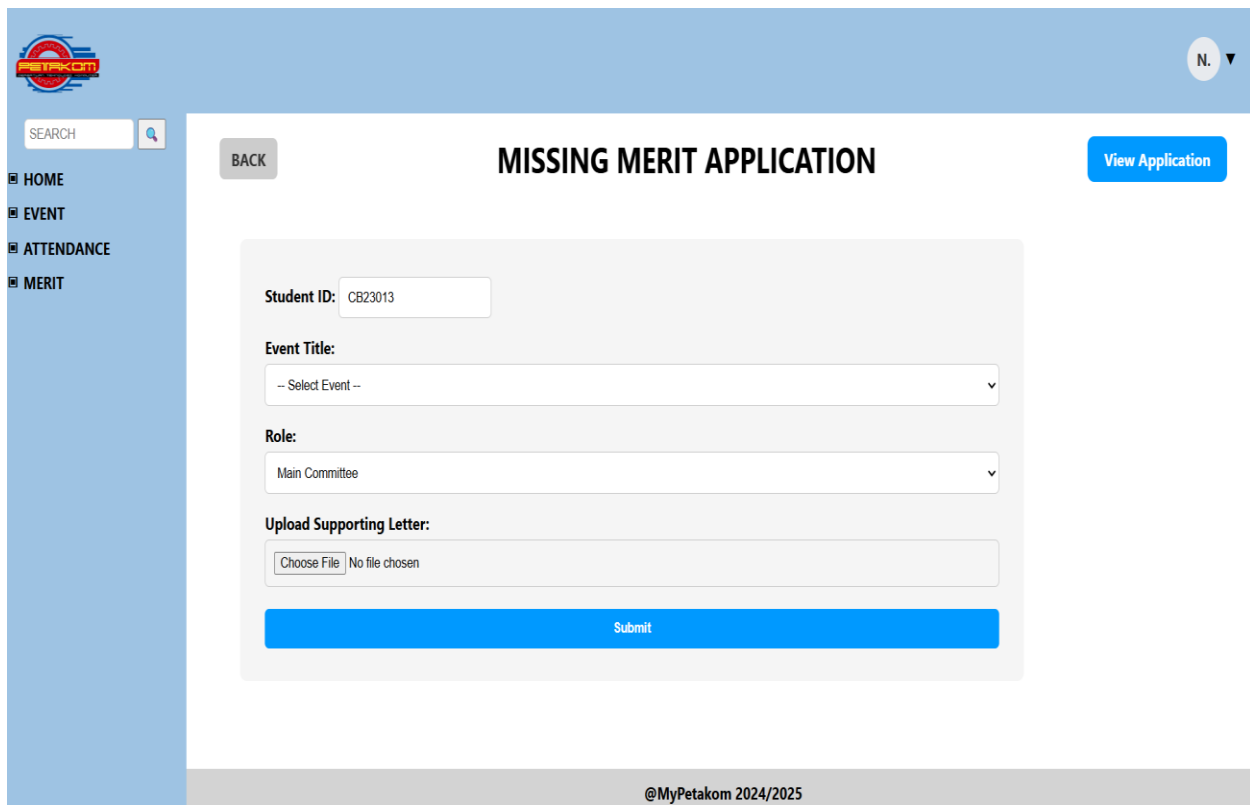
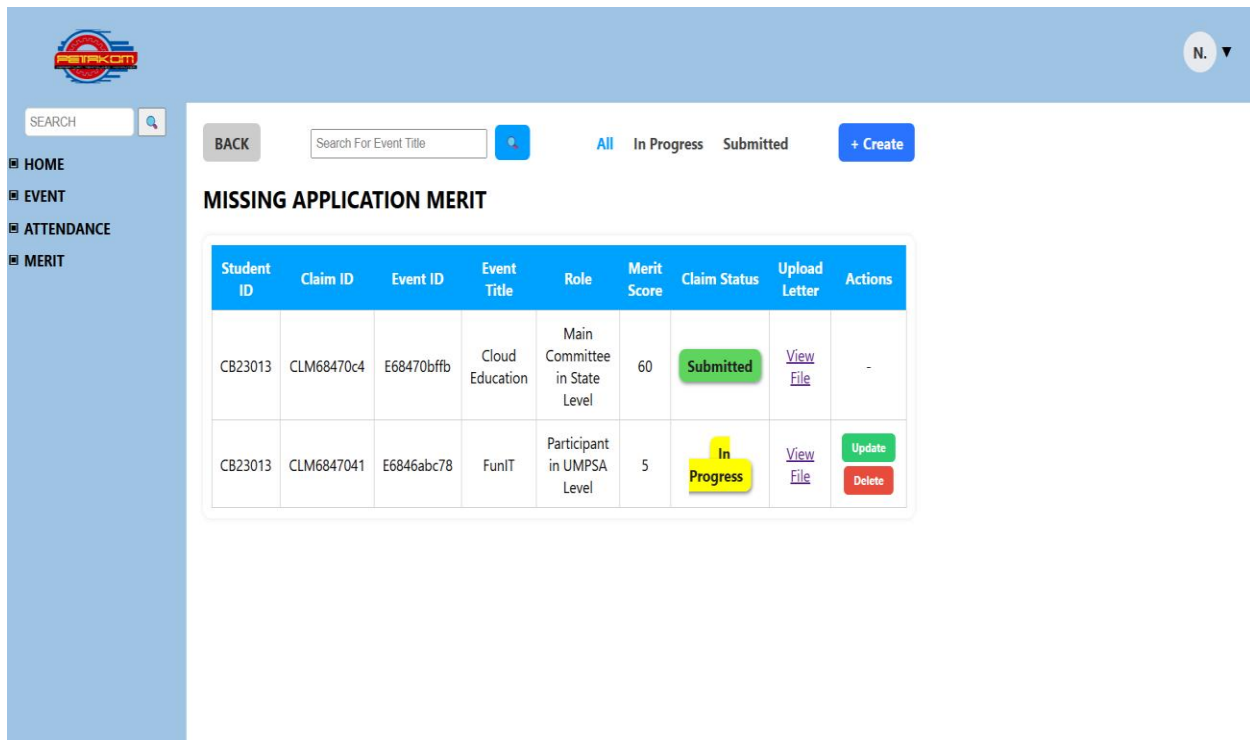


Figure 8.37 My Petakom Student Dashboard



The screenshot shows the 'MISSING MERIT APPLICATION' page. On the left is a sidebar with a logo and navigation links: HOME, EVENT, ATTENDANCE, and MERIT. The top header contains a search bar, a user profile icon with the letter 'N.', and a 'View Application' button. The main form area includes a 'BACK' button, a 'Student ID' field with the value 'CB23013', an 'Event Title' dropdown menu showing '-- Select Event --', a 'Role' dropdown menu showing 'Main Committee', and an 'Upload Supporting Letter' section with a 'Choose File' button and the text 'No file chosen'. A large blue 'Submit' button is at the bottom of the form. The footer of the page reads '@MyPetakom 2024/2025'.

Figure 8.38 Missing Merit Application page



The screenshot shows the 'MISSING APPLICATION MERIT' page. It features the same sidebar and header as Figure 8.38. Below the header, there is a 'BACK' button, a search bar labeled 'Search For Event Title', and filter buttons for 'All', 'In Progress', and 'Submitted'. A '+ Create' button is also present. The main content is a table with the following data:

Student ID	Claim ID	Event ID	Event Title	Role	Merit Score	Claim Status	Upload Letter	Actions
CB23013	CLM68470c4	E68470bffb	Cloud Education	Main Committee in State Level	60	Submitted	View File	
CB23013	CLM6847041	E6846abc78	FunIT	Participant in UMPSA Level	5	In Progress	View File	Update Delete

Figure 8.39 Display Missing Merit Application page



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

BACK

UPDATE MERIT APPLICATION

View Application

Student ID:

CB23013

Event Title:

FunIT

Role:

Main Committee

Upload Supporting Letter:

Choose File

No file chosen

SAVE

Figure 8.40 Update Missing Merit Application page



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

BACK

Search For Event Title

All

In Progress

Submitted

+ Create

MISSING APPLICATION MERIT

Student ID	Claim ID	Event ID	Event Title	Role	Merit Score	Claim Status	Upload Letter	Actions
CB23013	CLM68470c4	E68470bffb	Cloud Education	Main Committee in State Level	60	Submitted	View File	-
CB23013	CLM6847041	E6846abc78	FunIT	Participant in UMPSA Level	5	In Progress	View File	Update Delete

localhost says
Delete this merit claim?
OK Cancel

Show desk

Figure 8.41 Delete Missing Merit Application page



SEARCH



HOME

EVENT

ATTENDANCE

MERIT

Merit Application

Merit Summary

N.

BACK

QR Code

MERIT AWARDED

Student ID	Event ID	Event Title	Role	Merit Score
CB23013	E68470bffb	Cloud Education	Main Committee in State Level	60
CB23013	E68468c2de	Coding Marathon	Main Committee in UMPSA Level	30
CB23013	E6846ad305	Cooking contest	Participant in UMPSA Level	5
CB23013	E6846abc78	FunIT	Participant in UMPSA Level	5
CB23013	E6846ab7b7	Night Run	Committee in UMPSA Level	20
CB23013	E68478f552	Power Up Your Microsoft Skills	Participant in District Level	15
CB23013	E6846ac0b6	zombie run	Participant in UMPSA Level	5

Figure 8.42 Display the Merit Awarded page

Your Merit QR Code

Scan the QR code below to view and print your merit information.



Download QR

Print QR

Figure 8.43 QR Code Merit Awarded page



Student Merit Details

Name	SITI NAJWA BINTI BAKRI
Student ID	CB23013
Programme	BCS
Total Merits	140

Event Merit Breakdown

Event Title	Event Date	Role	Merit Score
Cloud Education	12 Jul 2025	Main Committee in State Level	60
Coding Marathon	26 Jun 2025	Main Committee in UMPSA Level	30
Cooking contest	16 Aug 2025	Participant in UMPSA Level	5
FunIT	17 Sep 2025	Participant in UMPSA Level	5
Night Run	13 Oct 2025	Committee in UMPSA Level	20
Power Up Your Microsoft Skills	06 Nov 2025	Participant in District Level	15
zombie run	28 Jun 2025	Participant in UMPSA Level	5

Print as PDF

Figure 8.44 Display the Student Merit Details page After Scan QR Code



SEARCH

HOME

EVENT

ATTENDANCE

MERIT

N. ▾

BACK

Change Merit Claim Status

Claim ID	Student ID	Event Title	Letter	Current Status	Change Status
CLM68470c4	CB23013	Cloud Education	View File	Submitted	Submitted ▾ Update
CLM6847041	CB23013	FunIT	View File	In Progress	In Progress ▾ Update

@MyPetakom 2024/2025

Show desktop

Figure 8.45 Petakom Advisor (Staff) Update Merit Claim Status page.

9.0 TESTING PLAN

This section outlines the detailed testing strategies and procedures used to ensure that the MyPetakom web application functions correctly and provides a smooth user experience. The testing plan covers tests that focus only on the functional part of the system such as:

1. Content testing
2. Interface testing
3. Usability testing
4. Component-level testing
5. Navigation testing

9.1 CONTENT TESTING

Content testing is done to uncover the error in the static and dynamic content of the MyPetakom system, where it is to find the syntactic errors, semantic errors and the errors in the organization or structure of content. Manual review by the developers of the system is done to ensure there are none of the errors mentioned.

For the static content, such as each user's dashboard, is done to ensure there is no incorrect grammar usage and outdated information. Next, for the dynamic content, it is tested to verify the database generated outputs, such as event details and student names, are pulled correctly from the database and presented clearly on the interface. Lastly, the media that is uploaded into the system, such as student cards and approval letters, are displayed correctly when retrieved from the database. All content is expected to be accurate and structurally consistent to allow better user experience.

9.2 INTERFACE TESTING

Interface testing is done to verify that the user interfaces are functional, consistent and visually pleasing following the related design rules. The developers of the system used walkthrough review to ensure the UI elements behave correctly and consistently.

The interactive elements such as buttons, menus, and QR codes are validated and noted on their behavior, whether it does what it is planned to do or not. The use case walkthrough also allows use

case navigation, that verifies the interface flow based on the user roles. Visual consistency such as fonts, layouts, colours and alignments are also evaluated.

9.3 USABILITY TESTING

Usability testing is done to evaluate how easily users can interact with the MyPetakom system and complete key tasks, by using the test categories such as interactivity, layout, readability and aesthetics. User Acceptance Testing (UAT) is done to test the categories mentioned.

Firstly, to test the interactivity of the system, the buttons, dropdown menus and menu items are tested by the real users of the system. This is to ensure whether the items mentioned are understandable and easy to use. Next, the layout is tested to verify whether the page structure and positioning of elements aid navigation or not. Moreover, readability tests the language, graphic representations and legible fonts, whether they are easily understandable to the users or not. Lastly, reviewing the overall aesthetic of the system allows the users to criticize whether the layout, colour and design lead to ease of use and comfortability.

9.4 COMPONENT-LEVEL TESTING

Component-level testing, also known as functional testing, focuses on uncovering errors in the MyPetakom system by testing each of the modules independently to ensure completeness of the system. Use case testing is done to test both success and error scenarios of the modules.

In the first module, Login and Manage Membership, the user authentication, session handling, membership registration (for Students) and the CRUD operations for user profile management are tested. The expected output for each test is as follows:

- User authentication: Users that have registered in the system, following their role (Student, Advisor, and Administrator), can access specific dashboards.
- Session handling: The session login and kills are maintained properly.
- Membership registration: Students are allowed to upload their student card, and are correctly stored and retrievable in the database.

- User profile management: The user profile for each user can be created, viewed, edited and deleted successfully by the Administrator.

In the second module, Manage Event Registration, the CRUD operations for the event handling, the CRUD operations for the committee assignment, merit application and the QR code generation are tested. The expected output for each test is as follows:

- Event handling: The Advisor must be able to create, view, update and delete events that they have created. The events created must be saved with valid details only.
- Committee assignment: The committee are correctly assigned and linked to events.
- Merit application: The Advisor must be able to apply for the merit of the events that they handle.
- QR code generation: QR code for the event details can be generated and retrieved correctly.

In the third module, Manage Event Attendance, the CRUD operations for the attendance slot creation, QR code generation, and student check-in validation are tested. The expected output for each test is as follows:

- Attendance slot creation: The Advisor must be able to create, view, update and delete attendance slots for the events that they have created.
- QR code generation: QR code generated will direct the Students to the correct attendance page.
- Student check-in validation: Only authorized Students can check-in. Attendance is recorded in the database without duplication.

In the fourth module, Manage Merit, the CRUD operations for the merit claim, status tracking, merit calculation logic and QR code generation are tested. The expected output for each test is as follows:

- Merit claim form: Students can create, view, edit and delete merit claim forms before submission for the events that they have attended.
- Status tracking: Advisors can approve the claims submitted and the data is reflected on the Student page.
- Merit calculation logic: System correctly calculates cumulative merits for each Student based on the event level and role.
- QR code generation: QR code generated links to the student merit summary. The merit summary generated is different for each Student as it is retrieved from the database.

9.5 NAVIGATION TESTING

Navigation testing is to ensure that the elements used for navigation in the MyPetakom system work efficiently and functionally. The mechanisms tested in this system are navigation links, redirects and internal search features.

For the navigation links, it is tested to ensure the internal and external links route to the correct pages. Redirect handling is done to validate error pages by requesting both correct and incorrect internal links and assess how the MyPetakom system handles each request. Lastly, the internal search features are tested to evaluate the search accuracy and completeness. The navigation of the system should be tested to allow logical and valid content.

10.0 DEPLOYMENT PLAN

The MyPetakom web application will be deployed using a structured strategy to ensure a smooth transition from development to production. This deployment plan follows the Agile methodology, emphasizing phased implementation, active stakeholder involvement, and iterative refinement. The goal is to deliver a reliable and fully functional solution for managing PETAKOM's administrative tasks such as membership registration, event coordination, attendance tracking, and merit management.

Before the system is deployed, thorough pre-deployment preparation will be carried out. The development team will ensure that all modules such as login authentication, membership verification, event and committee management, attendance QR functionality, and merit claims are fully tested and function according to system requirements. The development environment will mirror UMPSA's infrastructure as closely as possible to guarantee compatibility and minimize configuration issues.

A multi-environment deployment setup will be used, consisting of development, staging, and production environments. During each sprint, new features are developed and tested within the development environment. Once verified, the system will be deployed to the staging environment, which closely simulates the production setup. In the staging phase, realistic data and user scenarios will be used for final testing. Key stakeholders, including PETAKOM Coordinator and Event Advisors, will validate the application to ensure that all operational requirements are met. Upon successful verification and resolution of any issues, the application will be moved to the production environment, where it will become accessible to all users.

The deployment process will follow a staggered rollout strategy to reduce risks and ensure system stability. Initially, the system will be launched as a pilot version for a selected group of users, such as PETAKOM committee members. This pilot phase will provide valuable feedback on usability and performance. Based on this feedback, necessary improvements will be made before proceeding to a full-scale deployment. Once the system performs successfully during the pilot phase, it will be made available to all users, including students, staff, and coordinators at both Pekan and Gambang campuses.

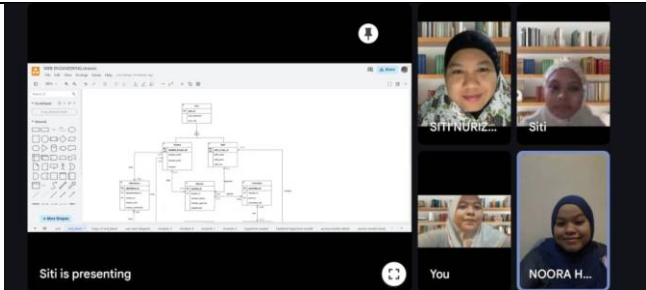
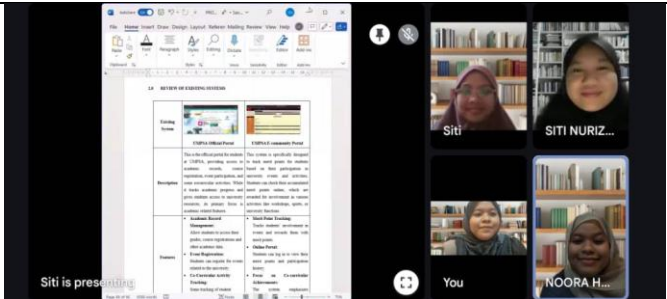
The MyPetakom system will be hosted using GitHub, with deployment handled through GitHub Pages. The final production-ready code will be pushed to the main branch of the GitHub repository, and the hosting service will be configured for auto-deployment. Database integration, HTTPS configuration, and live testing will be completed before the system goes live. Post-deployment, the development team will monitor performance, handle bug reports using GitHub Issues, and continue updating the application based on feedback through regular Agile sprints.

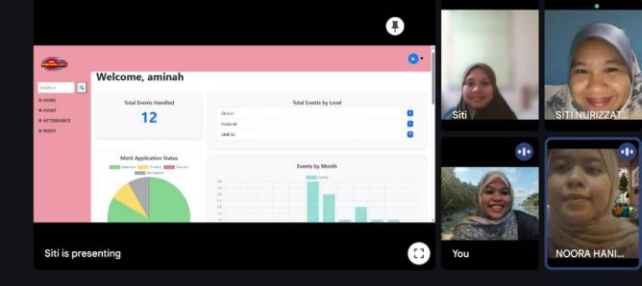
By following this deployment strategy, the MyPetakom system will be successfully launched as a robust and user-friendly platform, providing PETAKOM and UMPSA students with a seamless digital experience for managing co-curricular and event-related processes.

11.0 TASK DISTRIBUTION

NO	STUDENT NAME	TASK DISTRIBUTIONS & MODULES
1	SITI NURIZZATI BINTI ZABIDI (CA23075)	- MODULE 1 - Principle of Solution - Review of Existing Systems - ERD - Use Case Scenarios - Data design - Proposed Design - Interface Design
2	SITI NAJWA BINTI BAKRI (CB23013)	- MODULE 2 - ERD - Use Case Diagram - Use Case Scenarios - Access Model - Testing Plan - Proposed Design - Interface Design
3	NOORA HAIDAH BINTI AZMI (CB23095)	- MODULE 3 - ERD - Use Case Scenarios - Access Model - Testing Plan - Deployment Plan - Proposed Design

		- Interface Design
4	NOORA HANIM BINTI AZMI (CB23040)	- MODULE 4 - Project Management Framework - Project Requirements - ERD - Use Case Scenarios - Access Model - Hypertext Model - Proposed Design - Interface Design

Date	Progress	Reference
20/04/2025	- We distributed the tasks and modules that need to be done for this project. - We came up with the ERD for this project.	
28/04/2025	- After consultation with the lecturer, we gathered and redesigned the ERD.	

	- We finished the proposal for this project.	
02/06/2025	- We finished doing each other's modules and integrating them in one system.	 The screenshot shows a video conference interface. On the left, a presentation slide titled 'Welcome, aminah' is displayed. The slide features a sidebar with a search bar and a list of items. The main content area includes a 'Total Events Handled' section with a large number '12', a 'Total Events by Level' table, a 'Month Application Status' pie chart, and an 'Events by Month' bar chart. The status bar at the bottom of the presentation indicates 'Siti is presenting'. On the right side of the video conference, four participants are visible in a grid layout. Their names are displayed below their video feeds: 'Siti', 'SITI NURIZZA', 'You', and 'NOORA HANI...'. The 'You' participant is highlighted with a blue border.